FARM MANAGEMENT SYSTEM

A Project Report

BACHELOR OF COMPUTER APPLICATIONS

To The

Thiruvalluvar University, Serkkadu, Vellore – 632 115

By

Ms. Srinithi A(Register No:31823U09094)

Ms. Mythily N(Register No:31823U09062)

Ms. Mouwlika J R (Register No:31823U09059)

Ms. Sandhiya V (Register No:31823U09085)

Ms. Shamili Amritha T M(Register No:31823U09091)

MARUDHAR KESARI JAIN COLLEGE FOR WOMEN,
VANIYAMBADI

(AFFILIATED TO THIRUVALLUVAR UNIVERSITY)

Serkadu, Vellore – 632 115

1. INTRODUCTION

1.1 Project Overview :

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

1.2 Purpose :

The main purpose of this project is to:

- Digitize agricultural data management
- Improve data accuracy and accessibility
- Automate record handling
- Enable real-time monitoring
- Support data-driven farming decisions
- Enhance productivity and transparency

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Farmer	Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	Frustrated and confused
PS-2	Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient
PS-3	Buyer	Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied
PS-4	System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure

2.2 Empathy Map Canvas

Target Users:

- Farmers
- Farm Managers
- Buyers
- System Administrators

Empathy Analysis

- Think & Feel: Need accurate and timely farm data
- Hear: Feedback from agricultural officers
- See: Registers, paper files, spreadsheets
- Say & Do: Enter records manually
- Pain: Data loss, errors, delays
- Gain: Automation, fast reporting

2.3 Brainstorming

Brainstormed Ideas

- Manual farm records
- Excel-based management
- Web-based system
- Mobile farm app
- Salesforce-based FMS

Selected Idea

Salesforce-based Farm Management System.

3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

1. Login
2. Access Farm Management App
3. Enter Farm and Crop Data
4. Register Farmers and Buyers
5. Generate Reports
6. Analyze Results

7. Decision Making

3.2 Solution Requirements

FR No	Epic	Description
FR-1	Farmer Management	Manage farmer details such as name, contact information, farm location, and crops cultivated
FR-2	Crop Management	Maintain crop information including crop name, season, price, quantity, and availability
FR-3	Farm Management	Manage farm details such as farm size, soil type, location, and crops grown in the farm
FR-4	Buyer Management	Allow buyers to view available crops and purchase crops directly from farmers
FR-5	Crop Purchase	Enable buyers to request, confirm, and track crop purchases from farmers

Non-Functional Requirements

NFR-1: Usability

The system shall provide a simple and user-friendly interface that enables farmers and buyers to easily manage farm details and purchase crops without technical difficulties.

Navigation shall be organized through clear menus, forms, and dashboards.

Examples:

•Easy farmer registration form

•Simple crop listing interface

• Buyers can easily search available crops

NFR-2: Security

The system shall ensure the security of farmer and buyer information through authentication and access control. Sensitive data such as contact details and transaction records shall only be accessible to authorized users.

Examples:

•Only farmers can update their crop details

•Buyers can only view available crops

• Unauthorized users cannot modify farm data

NFR-3: Reliability

The system shall operate accurately and consistently to ensure correct crop listings, farm records, and purchase transactions without errors or duplication.

Examples:

•No duplicate crop entries

•Accurate crop availability updates

• Reliable purchase confirmation

NFR-4: Performance

The system shall provide fast response times when farmers update crop information and when buyers search or purchase crops.

Examples:

•Quick crop search by buyers

•Fast update of crop availability

• Smooth navigation between farm and crop records

NFR-5: Scalability

The system shall support increasing numbers of farmers, farms, crops, and buyers without affecting performance.

Examples:

•Add more farmers and farms easily

•Support larger crop databases

• Handle increasing buyer requests

NFR-6: Availability

The system shall remain available to farmers and buyers whenever they need to manage crops or purchase products.

Examples:

•Online access for farmers and buyers

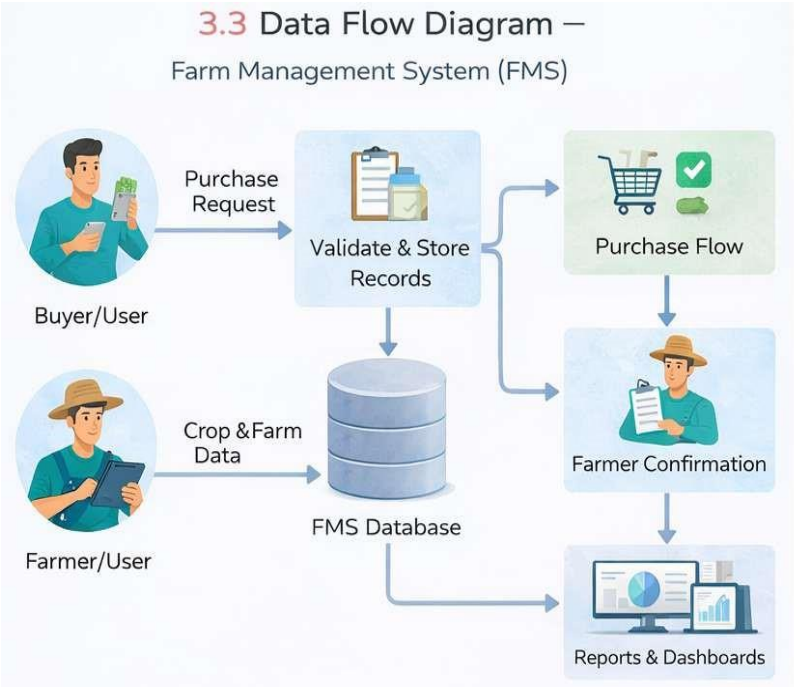
•Reliable during high demand periods

• Automatic backup of farm and crop data

3.3 Data Flow Diagram

- Farmer enters farm details and crop information
- System validates and stores farm and crop records
- Buyers search and view available crops
- Purchase requests are processed and stored in the system
- Reports and dashboards display farm production and crop sales data

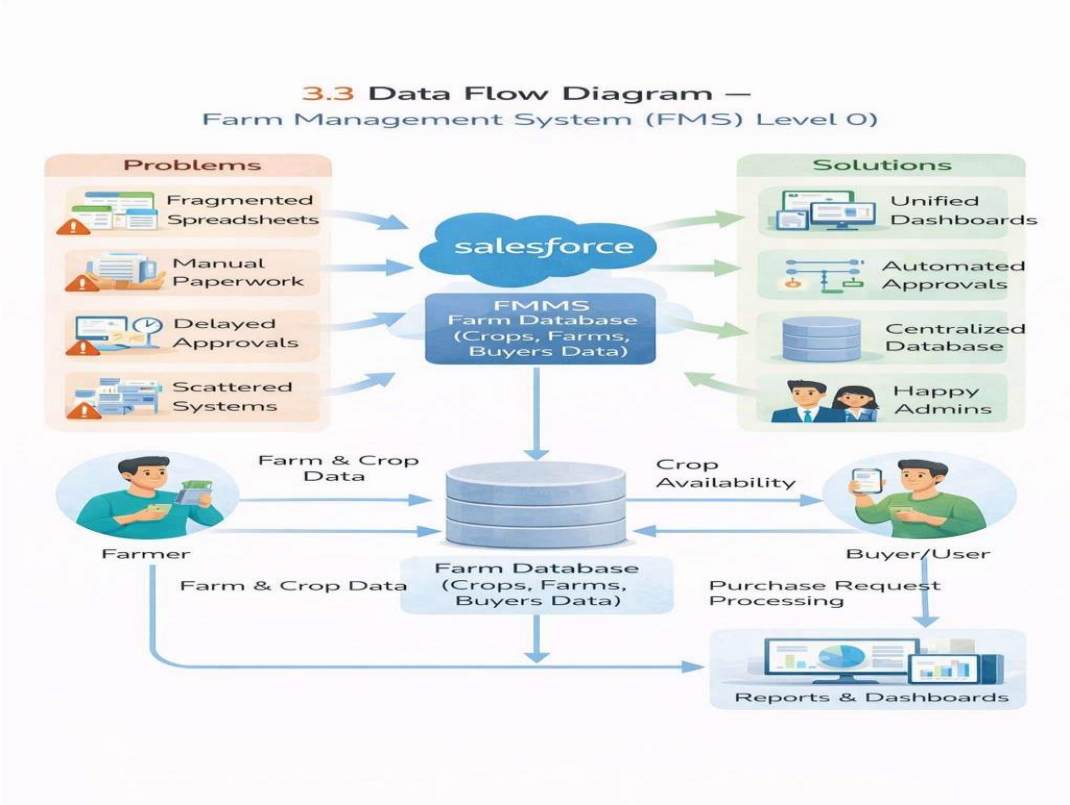
The system processes farm information, crop listings, and buyer purchase requests, validates the entered data, stores it in the database, and generates reports related to crop availability, farm production, and buyer transactions.



Layer	Technology
UI	Salesforce Lightning
Logic	Salesforce Flows
Database	Salessforce Objects
Automation	Validation Rules, Approval Process
Reporting	Reports & Dashboards

4.PROJECT DESIGN PHASE

4.1 Problem-Solution Fit



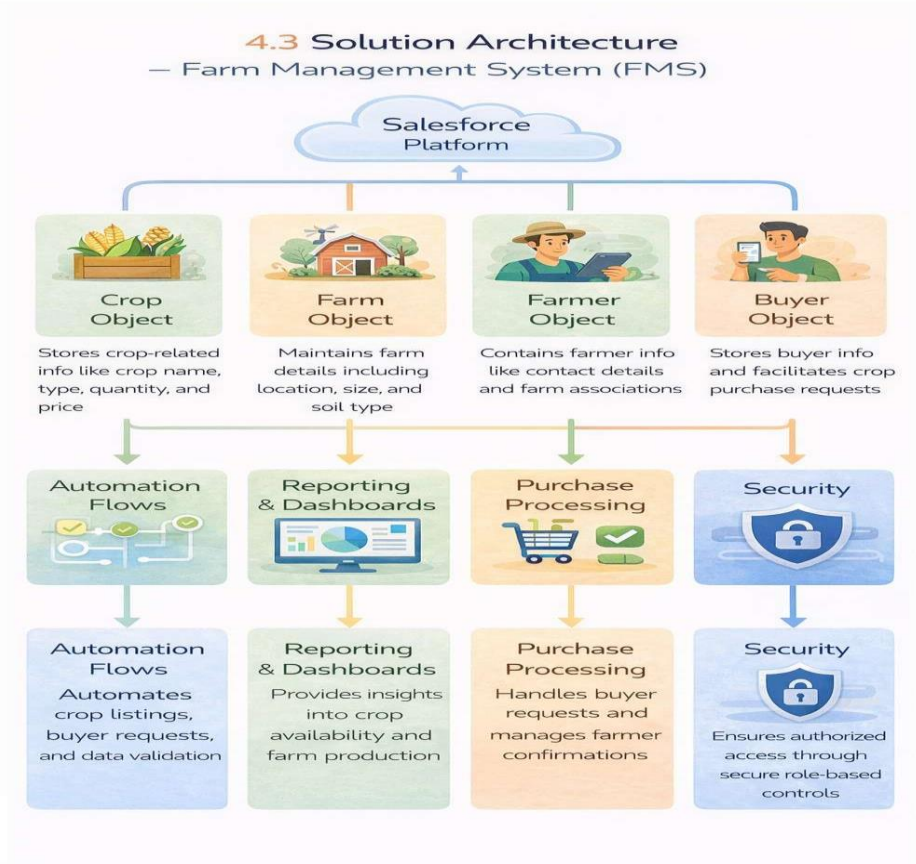
The Farm Management System (FMS) directly addresses fragmented agricultural management processes by aligning farmer and buyer activities, farm data management, and crop transactions using Salesforce-native capabilities to improve efficiency and data organization.

4.2 Proposed Solution

Functional Requirements

Parameter	Description
Problem	Manual farm record management and difficulty for buyers to find crops
Solution	Salesforce-based Farm Management System (FMS)
Novelty	Centralized platform connecting farmers and buyers with automated crop management
Social Impact	Supports farmers by improving crop visibility and enabling direct buyer access

Revenue Model	Platform deployment for agricultural organizations or farmer cooperatives
Scalability	Supports multiple farms, crops, farmers, and buyers across different regions



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

Sprint	Functional Requirement (Epic)	User Story No	User Story / Task	Story Points	Priority	Team Members

Sprint-1	Setup	USN-1	Create a Salesforce developer account to configure the Farm Management System.	1	High	Member1
Sprint-1	Data Modeling	USN-2	Create custom objects for Crop, Farm, Farmer, and Buyer.	3	High	Member2
Sprint-2	Data Modeling	USN-3	Create tabs for Crop, Farm, Farmer, and Buyer for easy navigation.	3	High	Member3
Sprint-2	Data Modeling	USN-4	Add fields such as crop name, farm location, farmer details, and crop price.	5	High	Member4
Sprint-2	Data Modeling	USN-5	Create relationships between Crop, Farm, Farmer, and Buyer objects.	3	High	Member4
Sprint-2	Data Modeling	USN-6	Define relationships between farms and crops.	5	Medium	Member1
Sprint-2	Data Modeling	USN-7	Customize page layouts for farm and crop records.	5	High	Member2
Sprint-3	Automation	USN-8	Validate crop and farm data during entry.	3	High	Member4
Sprint-3	Automation	USN-9	Automate buyer purchase request handling.	3	High	Member4
Sprint-3	Automation	USN-10	Create approval process for crop purchase requests.	5	High	Member1
Sprint-4	Security	USN-11	Configure role-based access for farmers and buyers.	5	High	Member2
Sprint-5	Reports	USN-12	Generate reports for crops, farms, and purchases.	4	High	Member3

Sprint-6	Dashboards	USN-13	Create dashboards for crop production and buyer requests.	4	Medium	Member4

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	2 Days	27 Feb 2026	01 Mar 2026	20	08 Mar 2026
Sprint-2	20	2 Days	03 Mar 2026	03 Mar 2026		
Sprint-3	20	2 Days	04 Mar 2026	05 Mar 2026		
Sprint-4	20	2 Days	06 Mar 2026	07 Mar 2026		
Sprint-5	20	2 Days	08 Mar 2026	09 Mar 2026		

6. Project Development Phase:

Project Flow:

Milestone 1: Salesforce Developer Account Creation

Milestone 2: Custom Object Creation (Farm,Farmer,Buyer,Crop)

Milestone 3: Tabs Creation

Milestone 4: Lightning App Creation

Milestone 5: Fields Creation

Milestone 6: Create Relationships

Milestone 7: Validation Rules

Milestone 8: Reports

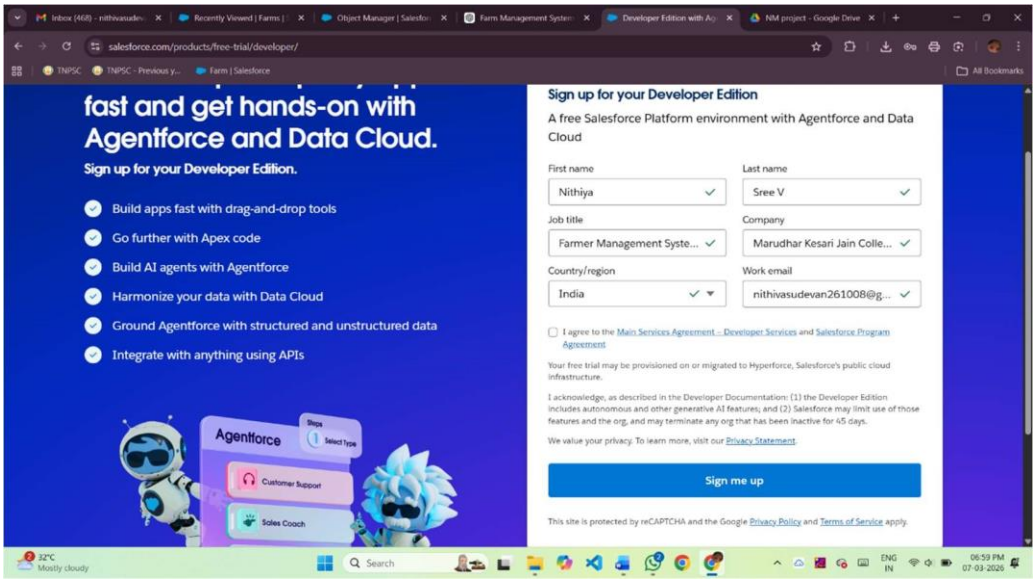
Milestone 9: Dashboards

Milestone 10: Conclusion

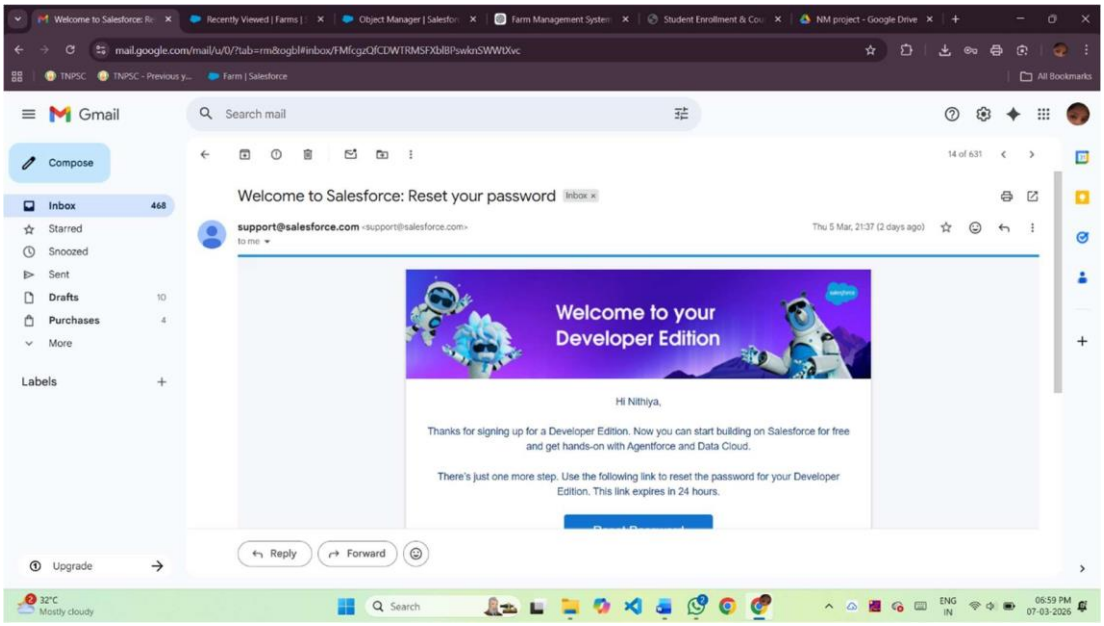
Implement:

Milestone 1-Salesforce :

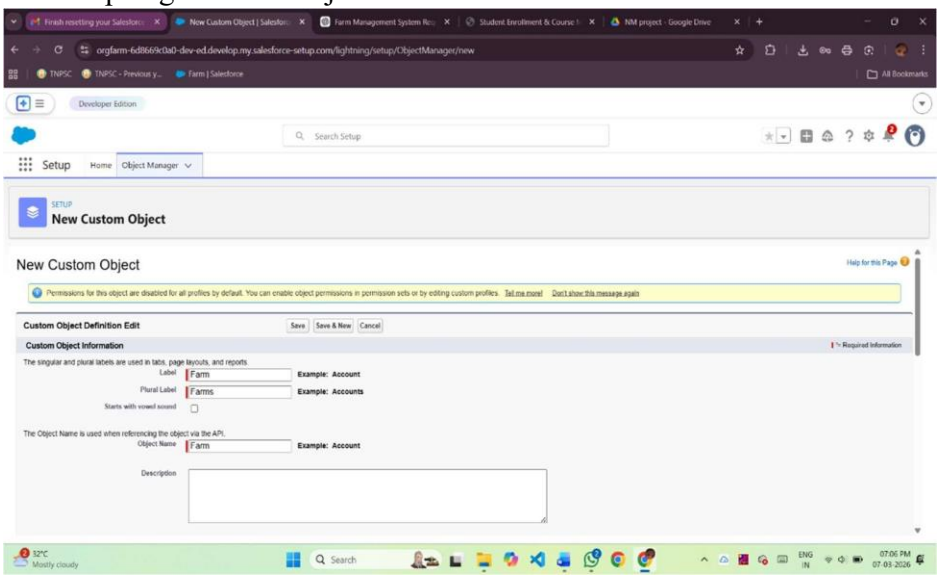
1. I Created a new account in salesforce as Nithiya Sree



2. As I logged in the first page

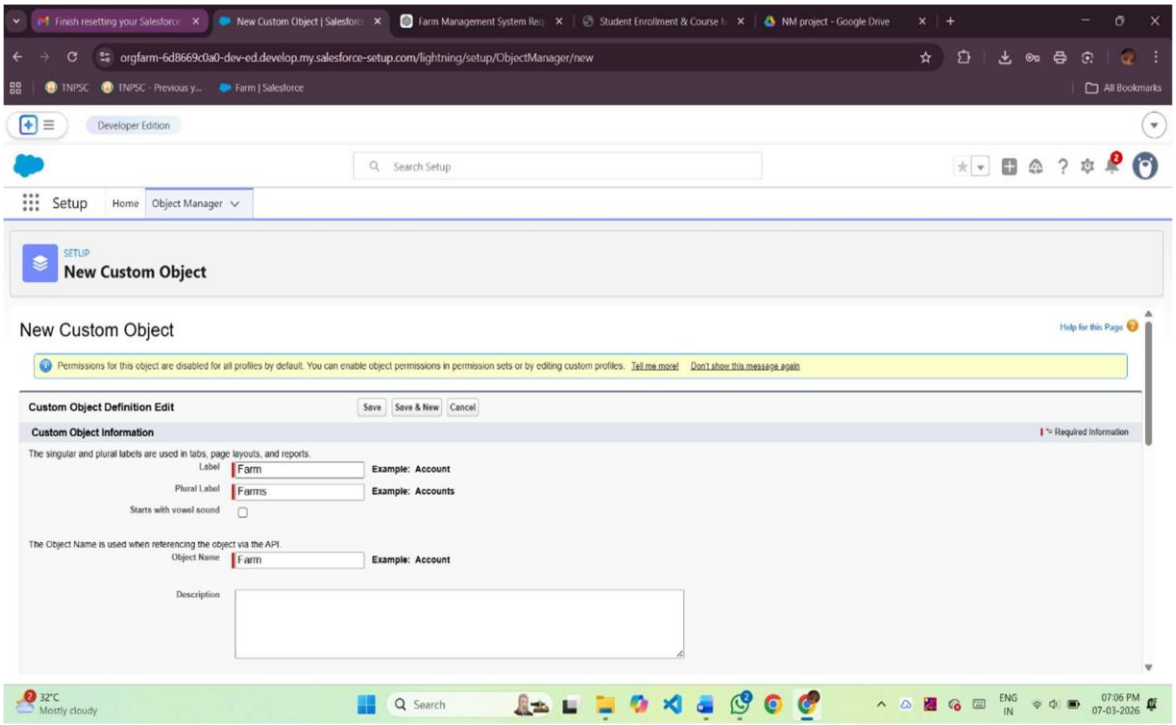


3. The Setup Page of Our Project



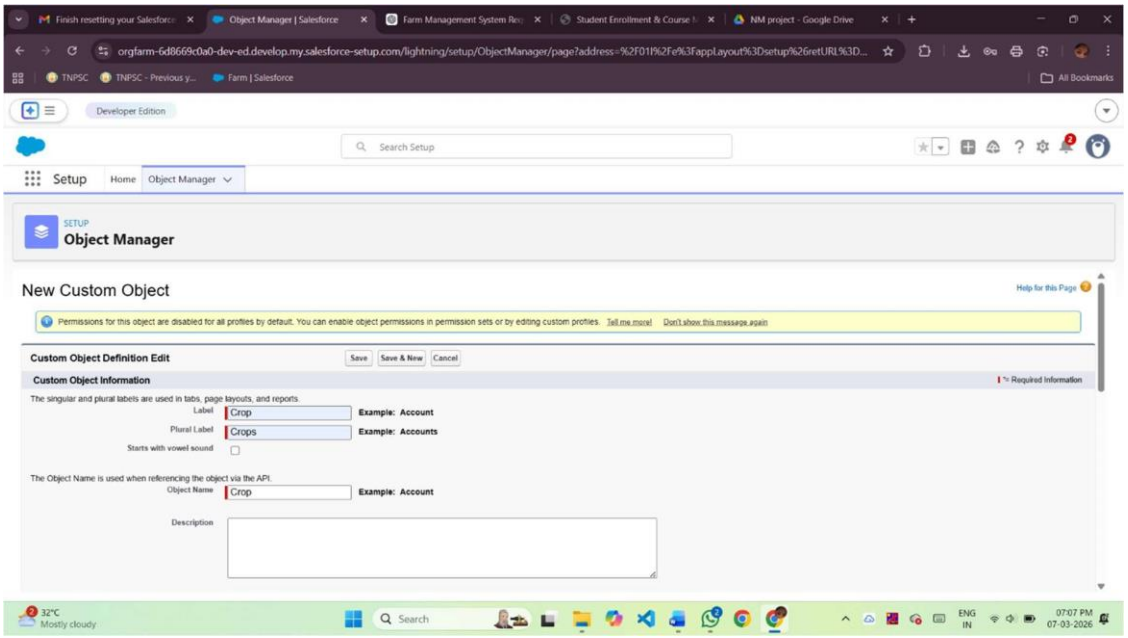
Milestone 2 - Object Creation:

1.Farm Object:

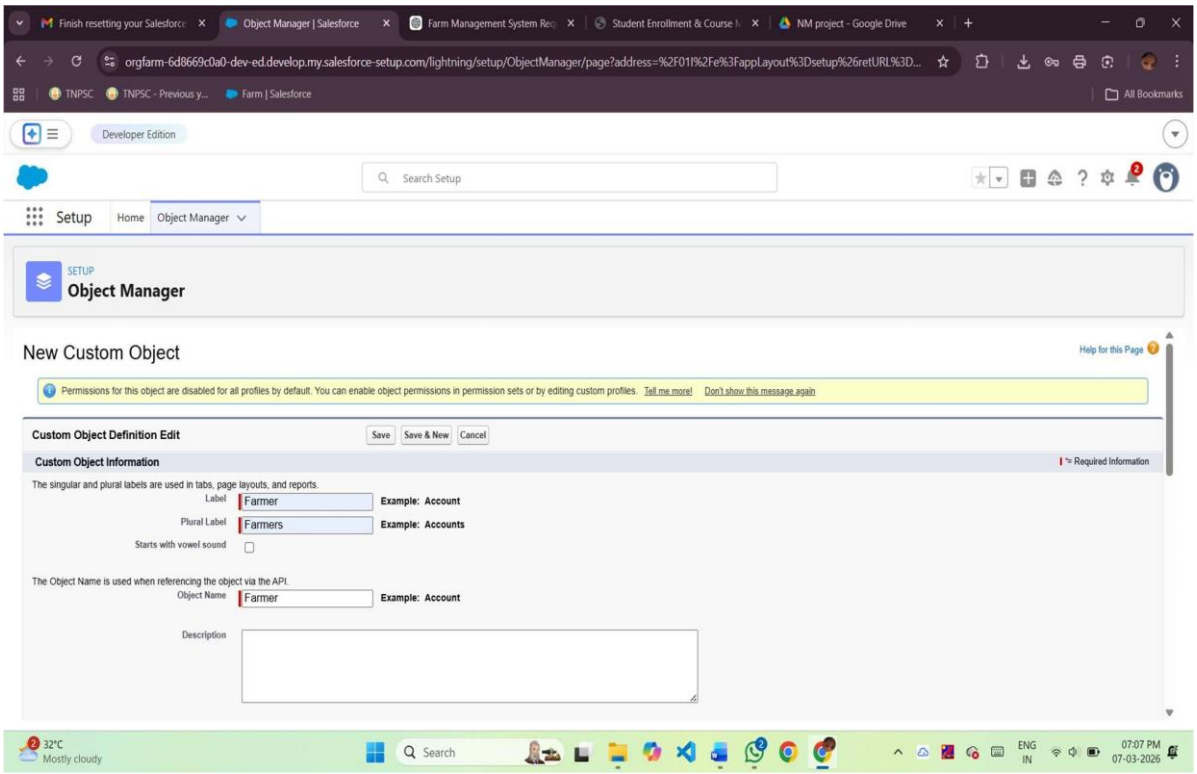


2.Crop Object:

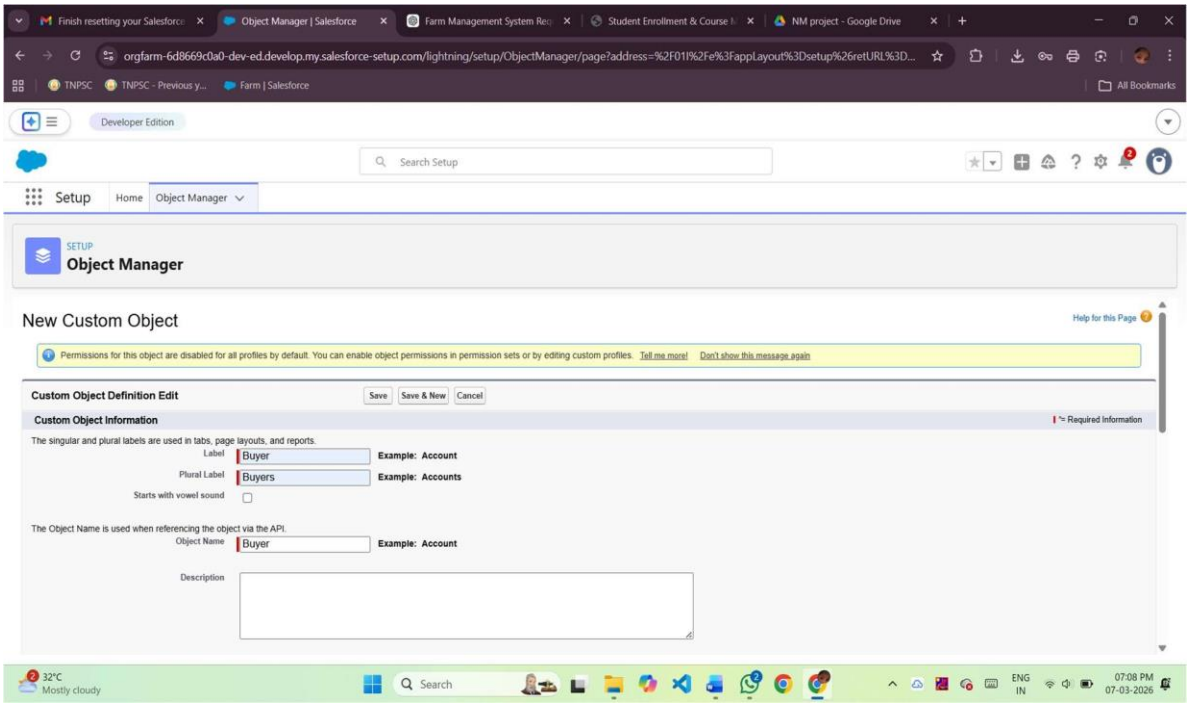
:



3.Farmer Object:



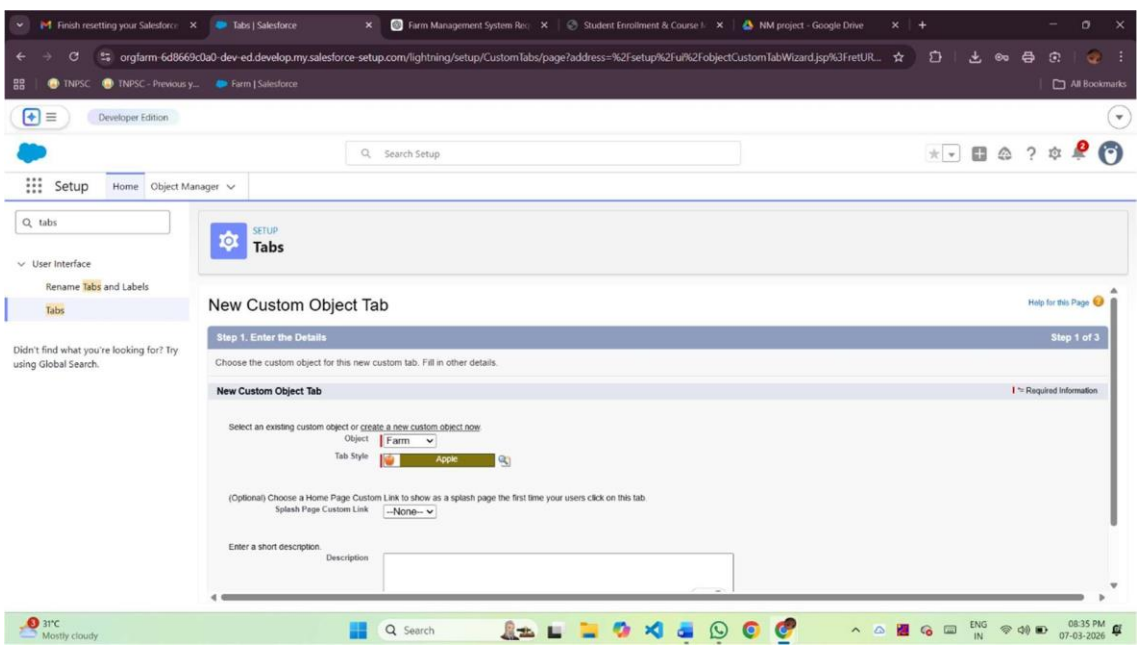
4.Buyer Object:



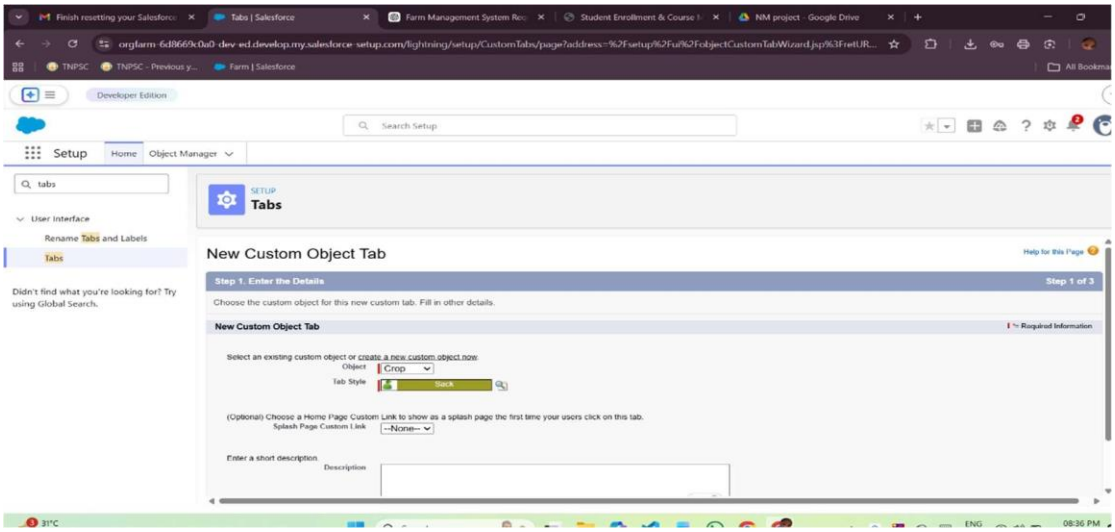
Milestone 3 – Tabs:

Custom Tab Creation:

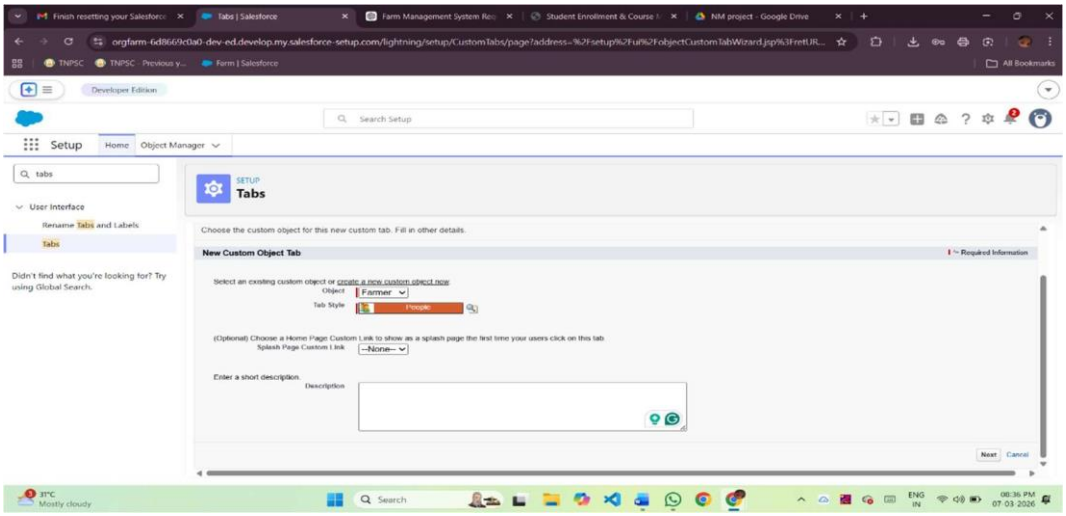
1.Farm Tab:



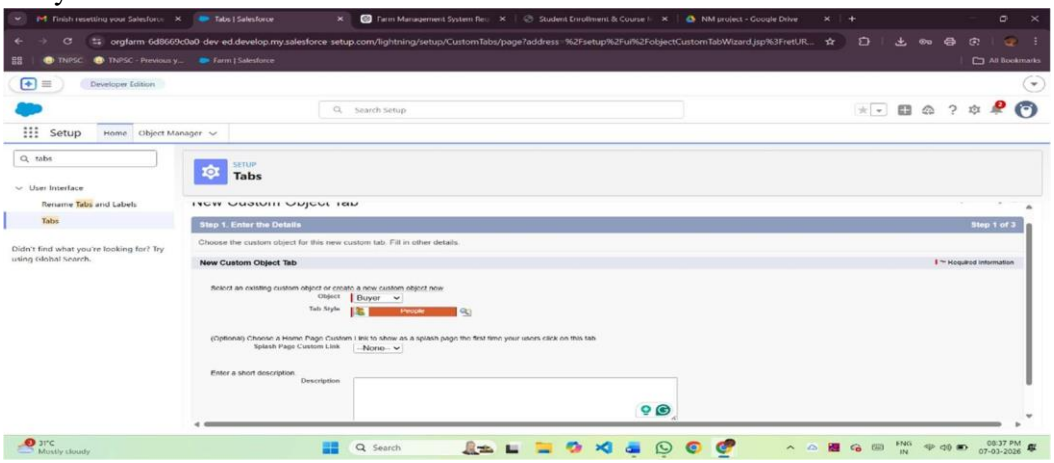
2.Crop Tab:



3.Farmer Tab:



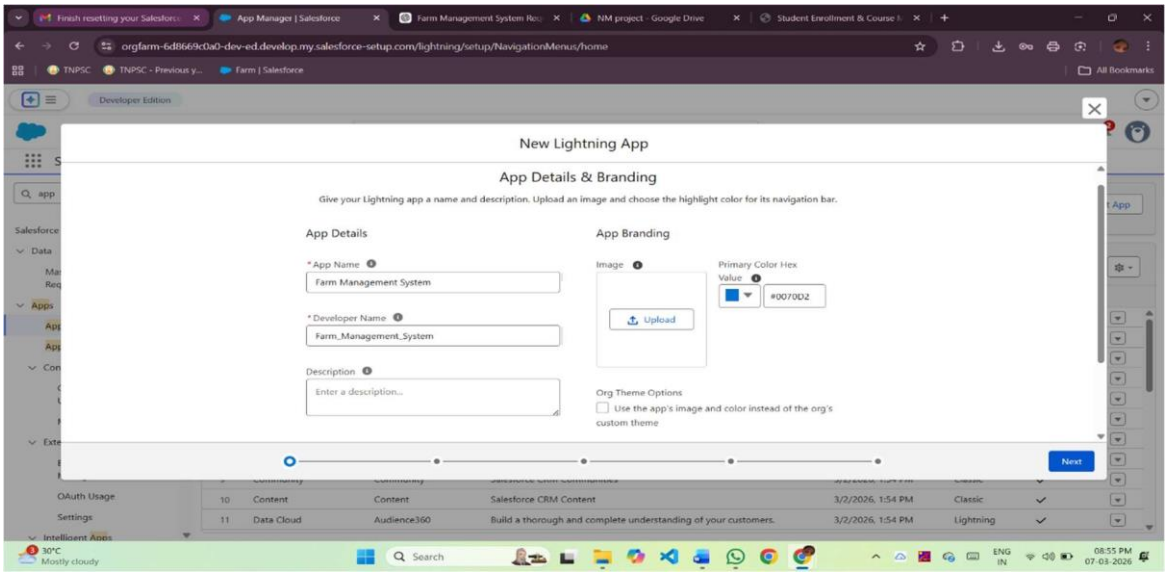
4.Buyer Tab:



Milestone 4 - The Lightning App

1 : Create a lightning app:

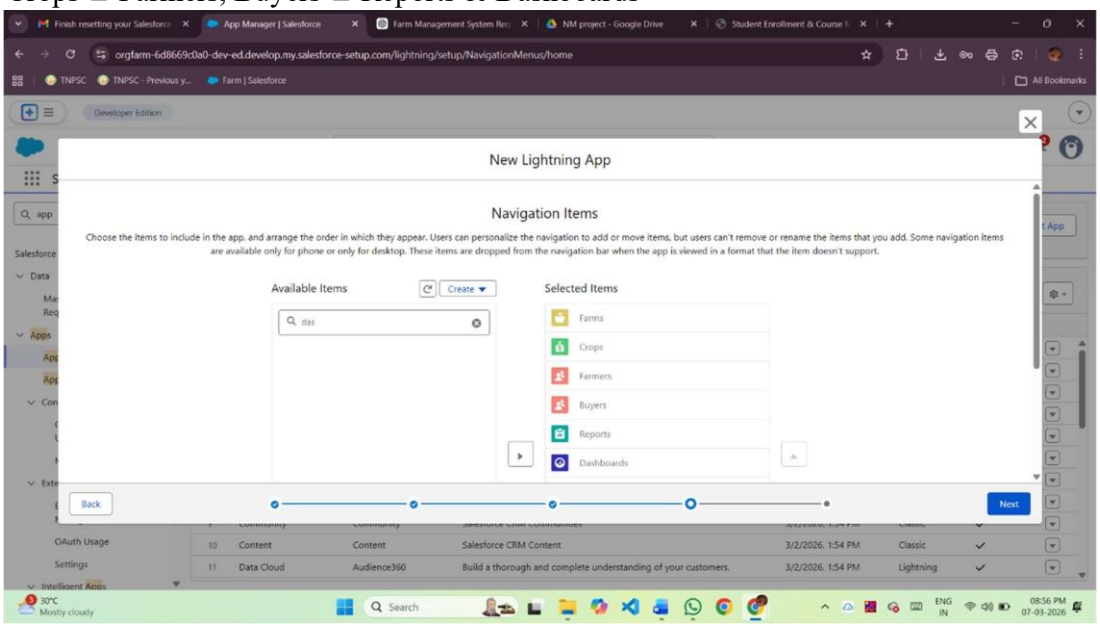
1. Go to setup page → search “app manager” in quick find → select “app manager”→ click on New lightning App and create app as Farm Management System



2. Navigation items

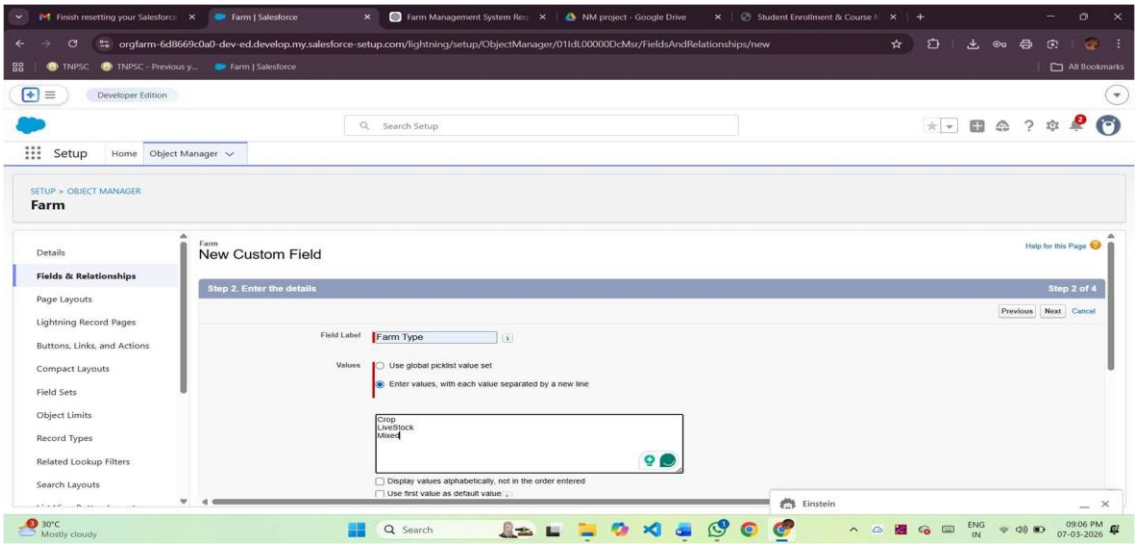
Select the created Custom Objects and required standard objects

- Farms, Crops
- Farmers, Buyers
- Reports & Dashboards



Milestone 5 : Fields & Relationships

1 : Create Custom fields for Farm Object:



1.Next we Create another Picklist Field Enter field label (Soil Quality) enter values (Organic, Cover Crops, Composting)

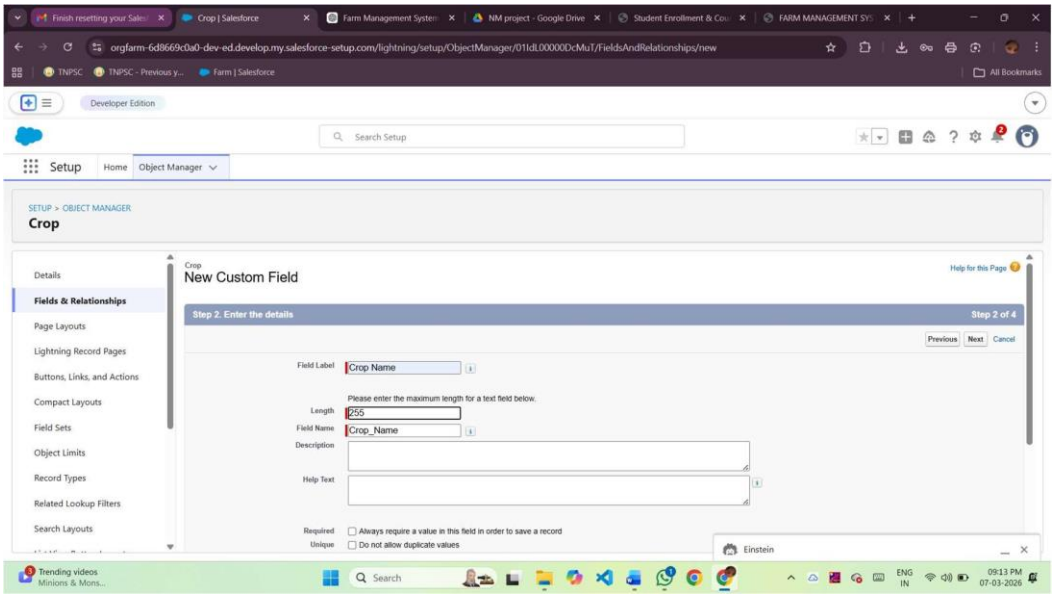
2.Again we Create Picklist field, Field Label (Irrigation type), Enter Values (Sprinkler irrigation, Drip irrigation, localized irrigation)

3.Choose data type as a text, enter field label(Location)

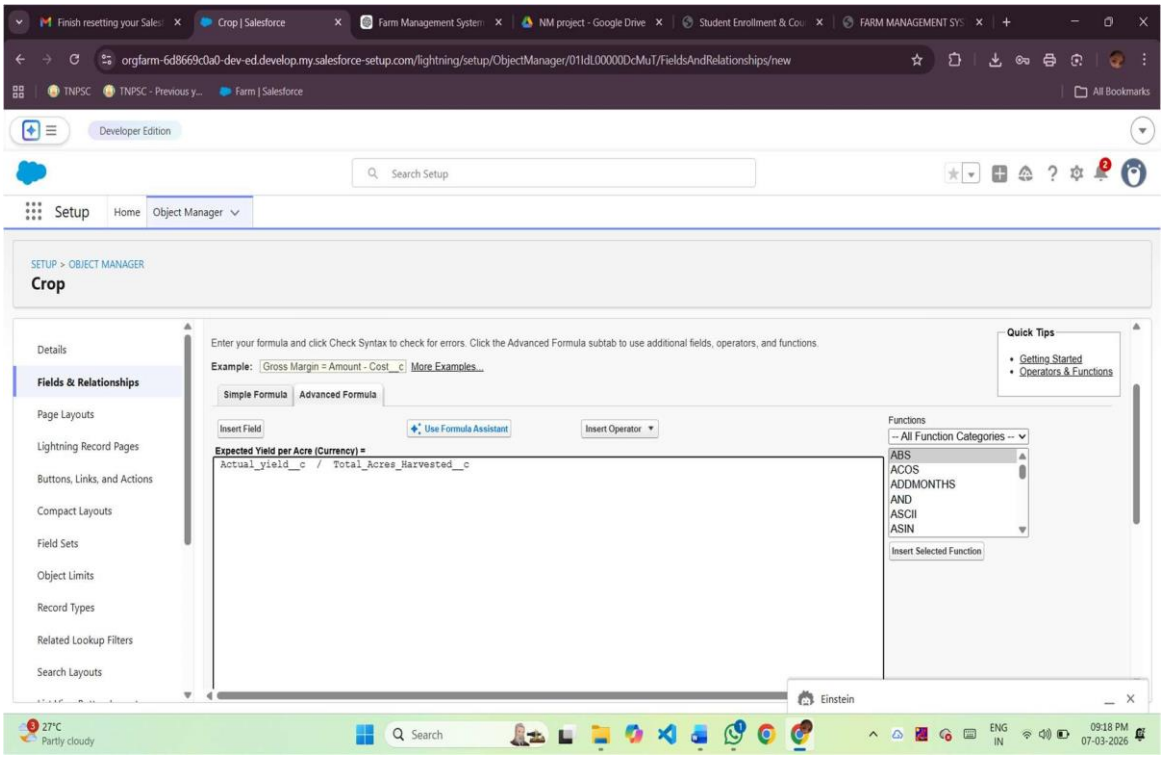
4.Choose data type as text, enter field label (size)

5.Create a Master Detail relationship with Farmer Object.

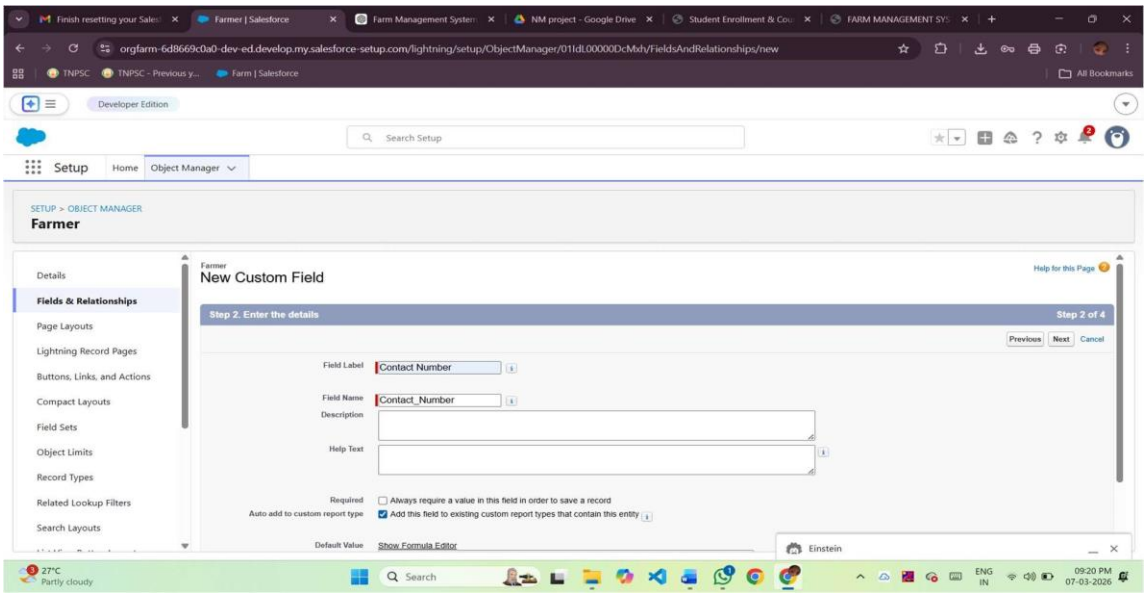
2.Create Custom fields for Crop Object:

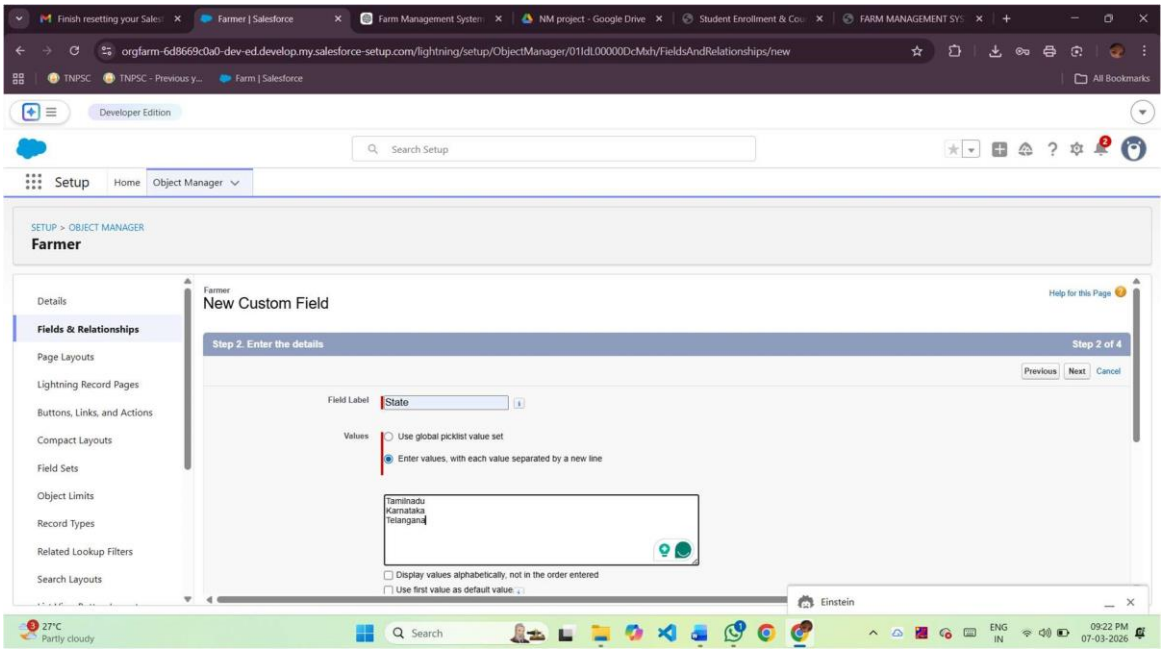


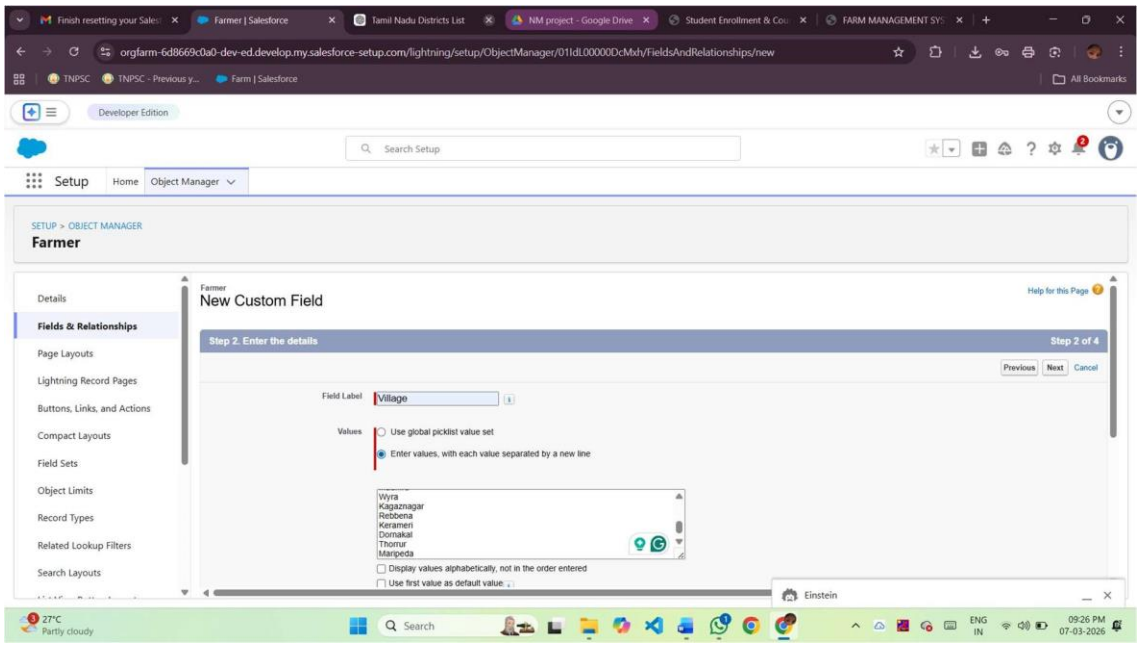
2.1 Formula in crop object:



3.Create Custom Field for Farmer Object:







Creating Field Dependencies:

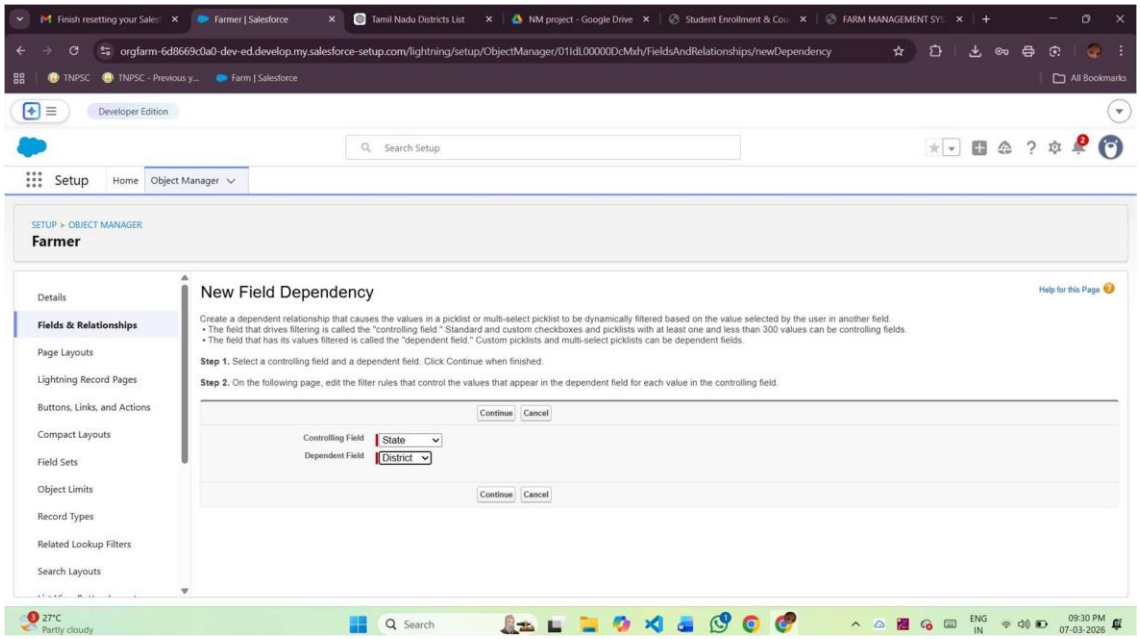
1.Now create Field Dependency For District field, Open Object Manager ,Choose Farmer object ,select Fields & Relationships, On Right side We have Field Dependency option ,select that option. 2.Click new, Select Controlling Field as State ,Dependent Field As District ,Click on Continue.

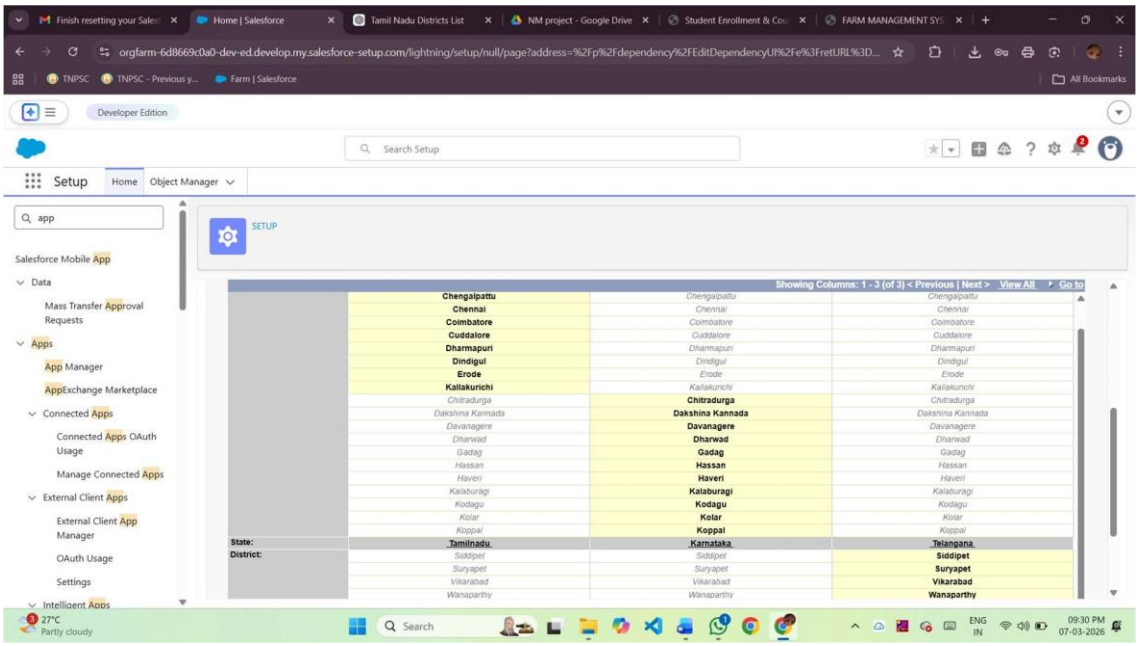
3. Select the appropriate Value in each column by double-clicking them.

4. For Ex. Telangana - YadadriBhongir.

5.Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its city.

6.Click Preview, then test the dependency by selecting a different State and viewing the associate Values available for Particular state.





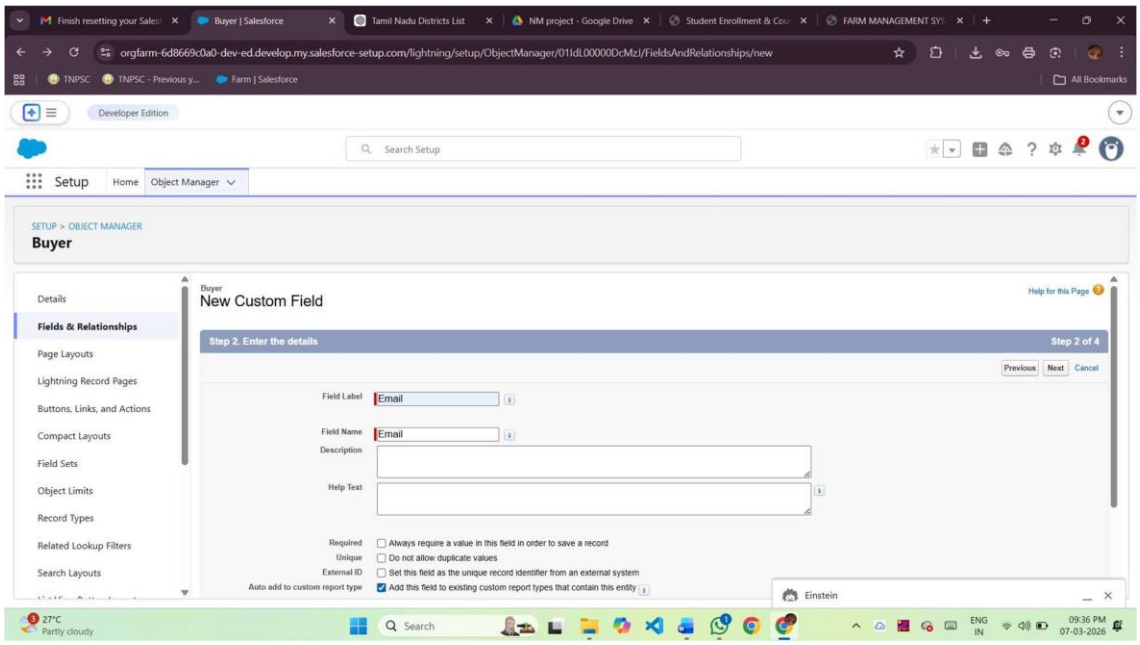
4.Create Custom Field for Buyer Object:

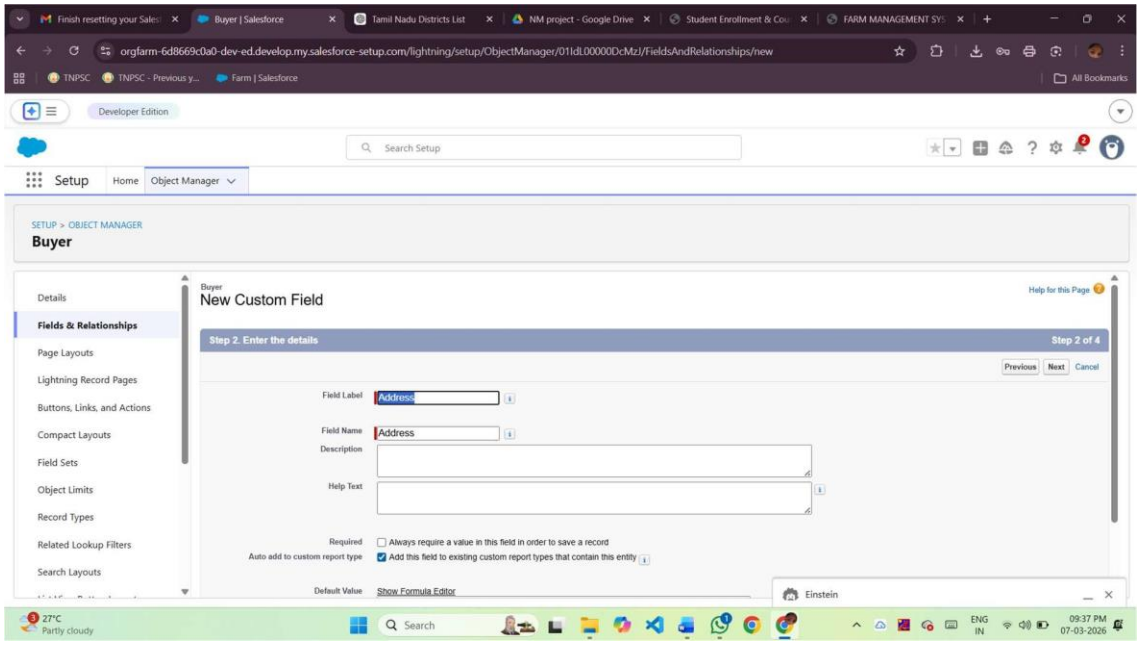
1.Select Data type as a Phone, Enter Field Label (Contact), Save.

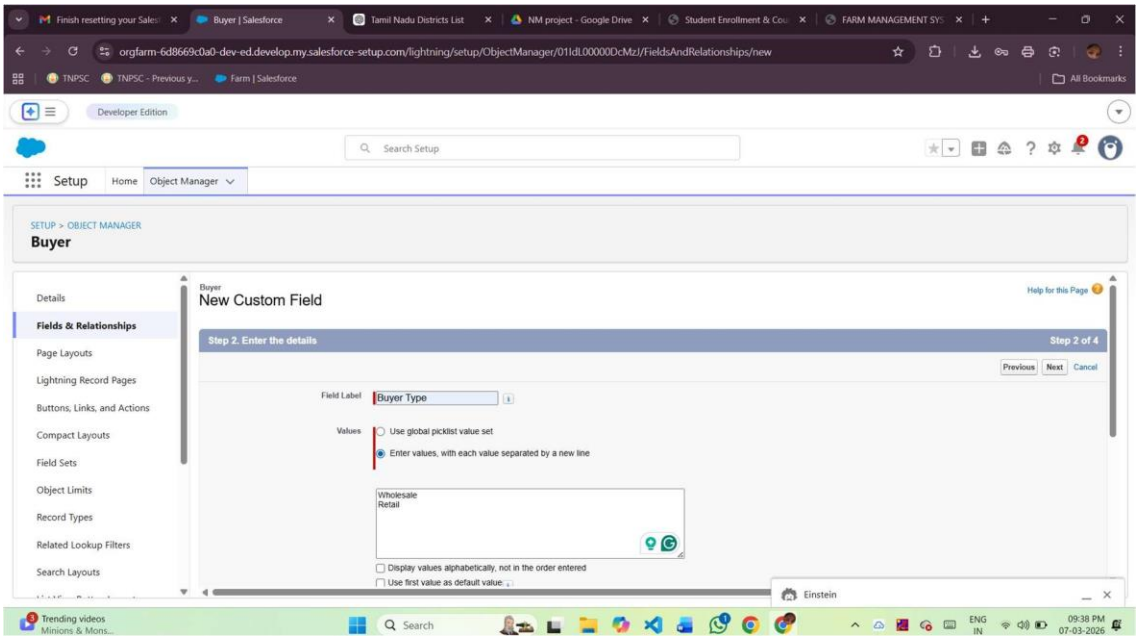
2.Select Data type as an Email, Enter Field Label (Email), Next save.

3.Select Data type Text area, Enter Field Label (Address), Save it.

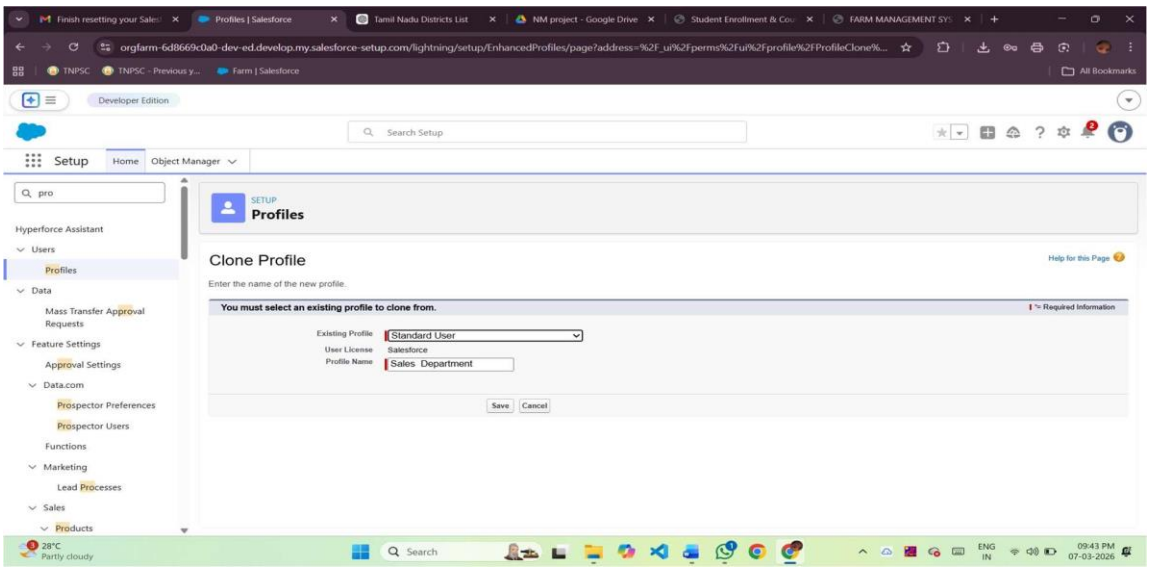
4.Select Data type Picklist, Field Label (Buyer Type),Enter values (Wholesale, Retail),Click Save.

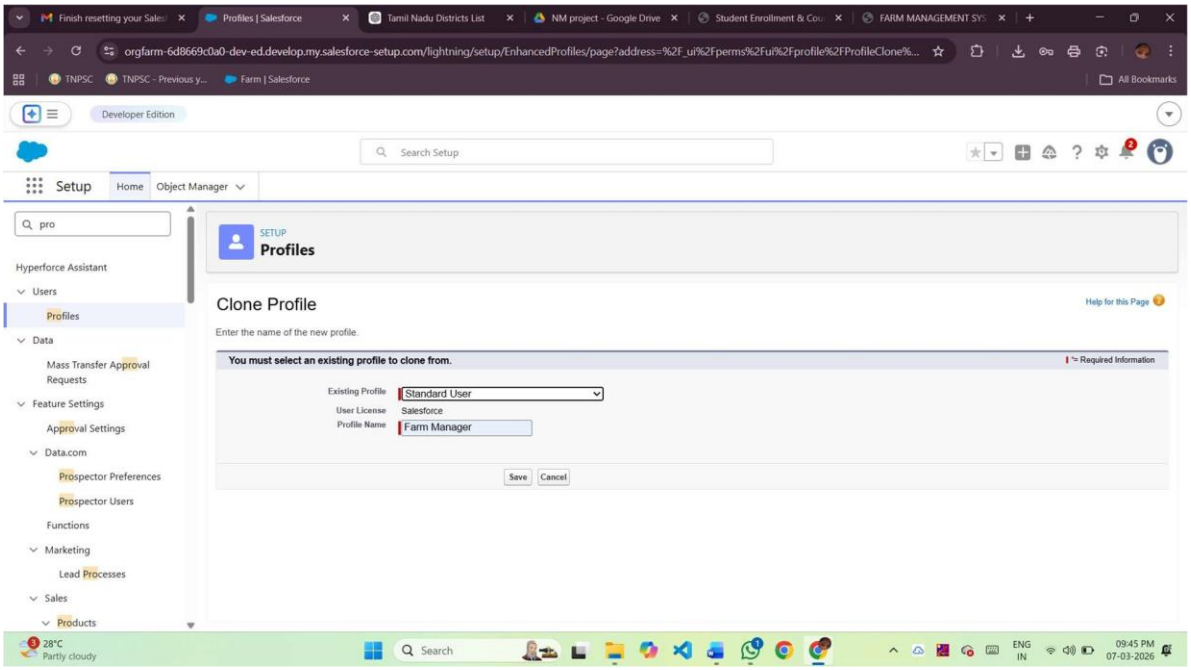




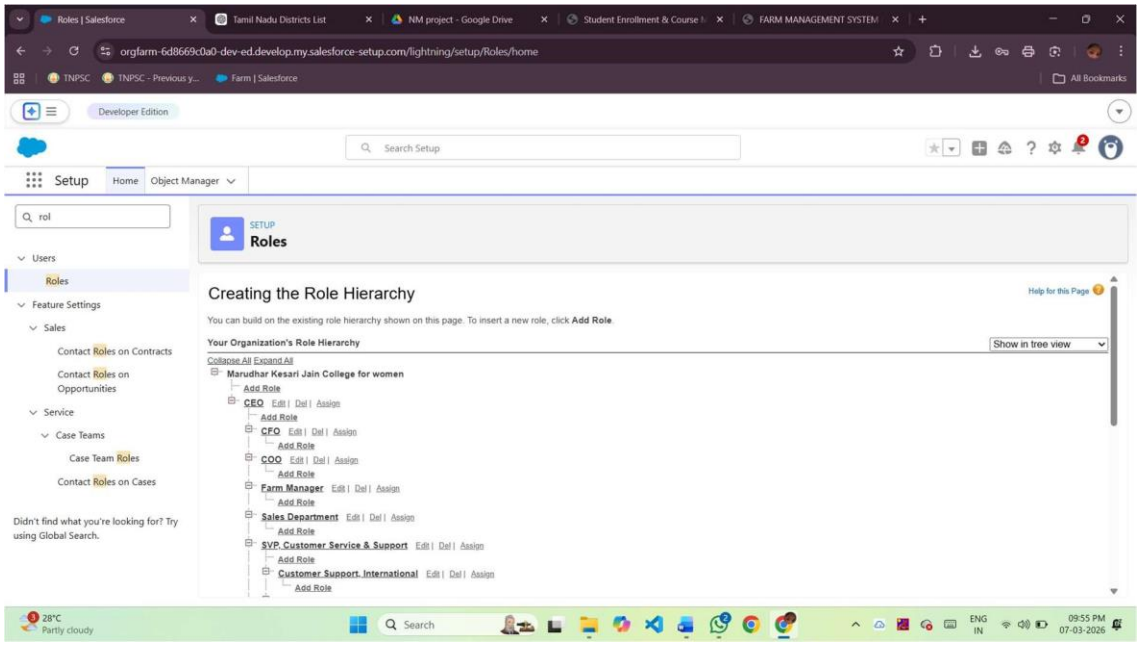


Mile Stone 6: Profile

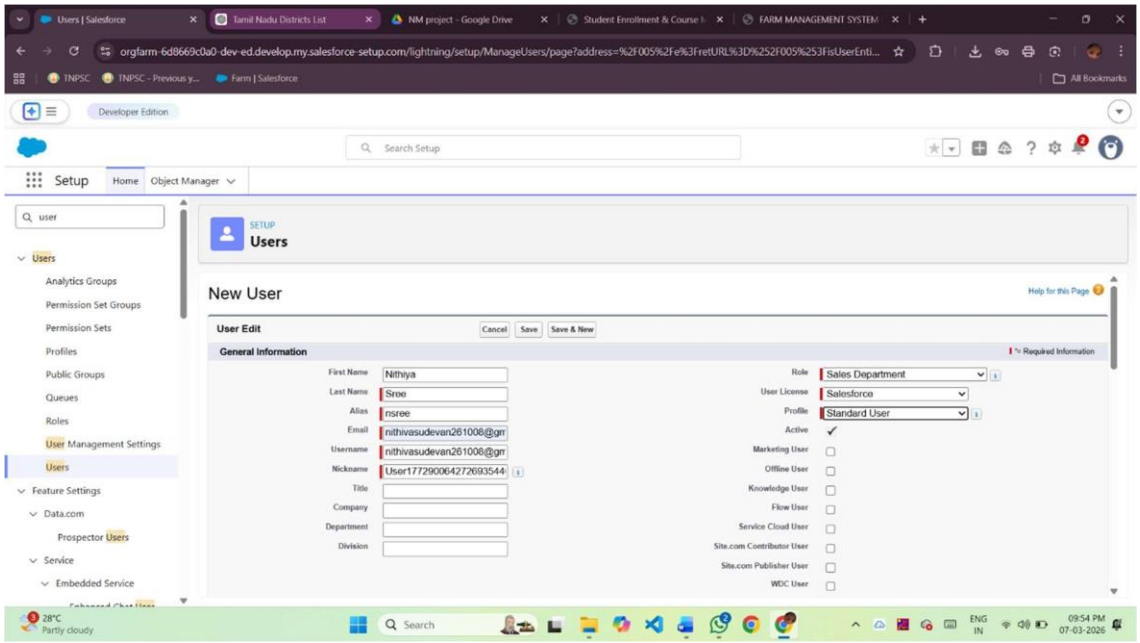




Milestone 7: Roles:



Milestone 8: Users

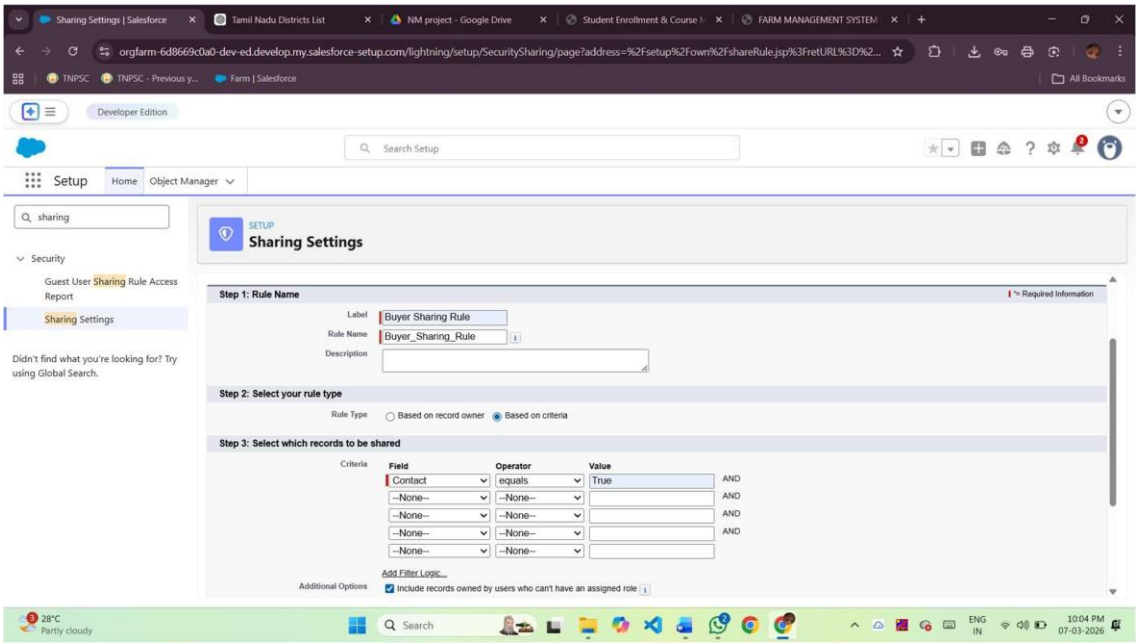


Milestone 9: Sharing Settings

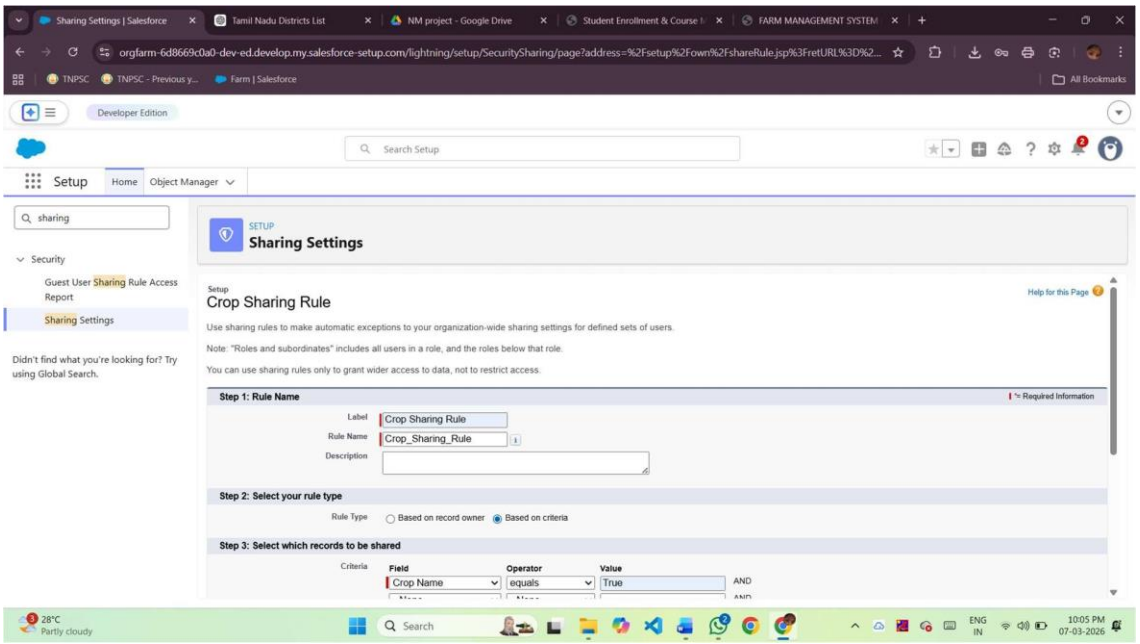
Create a sharing rule

1) Goto Sharing Settings, which can be found under the Quick Find section.
2) Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.
3) Add the label of the sharing rule you want to make.
4) Select your rule type based on the criteria.
5) Select the field can join immediately check field from the Crop object.
6) Select the Crop as equal and value is True.
7) And in selecting the users to share with the section select roles and in that select Farm Manager.
8) And in the section of select the level of access for the users give the access Read/Write.
9) And save the rule.

1.Buying Sharing setting:



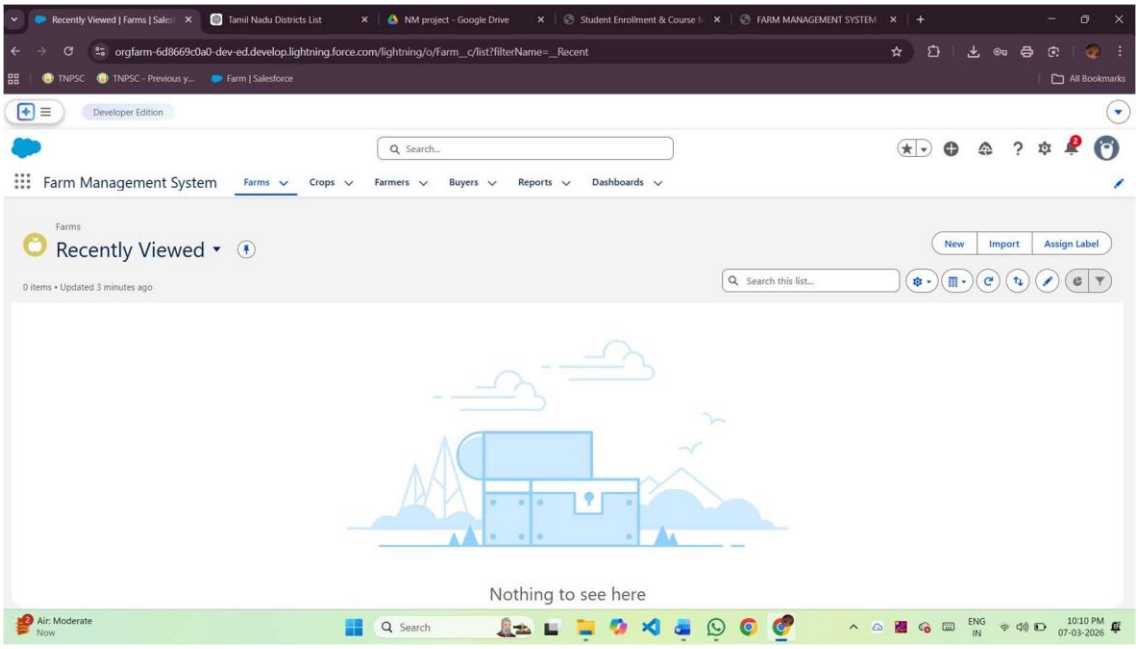
2.Crop Sharing setting:



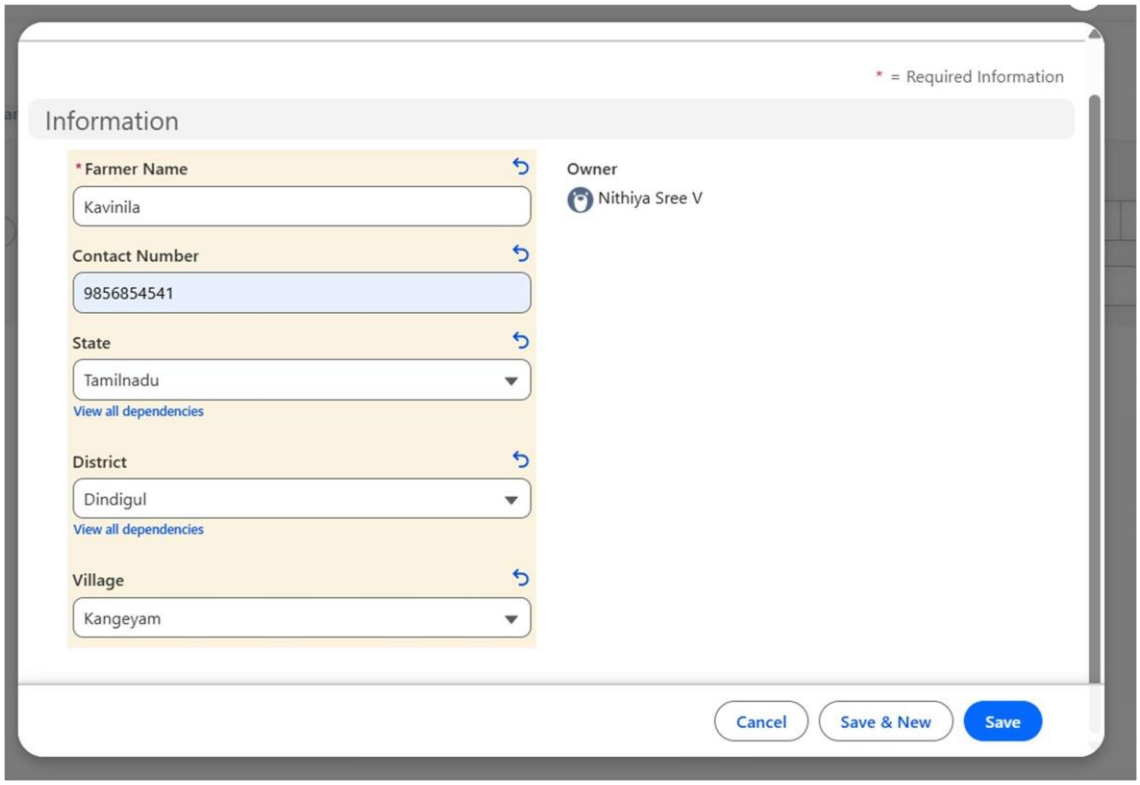
Milestone 10: App Launcher

Activity 1:Create a Record(Farmer)

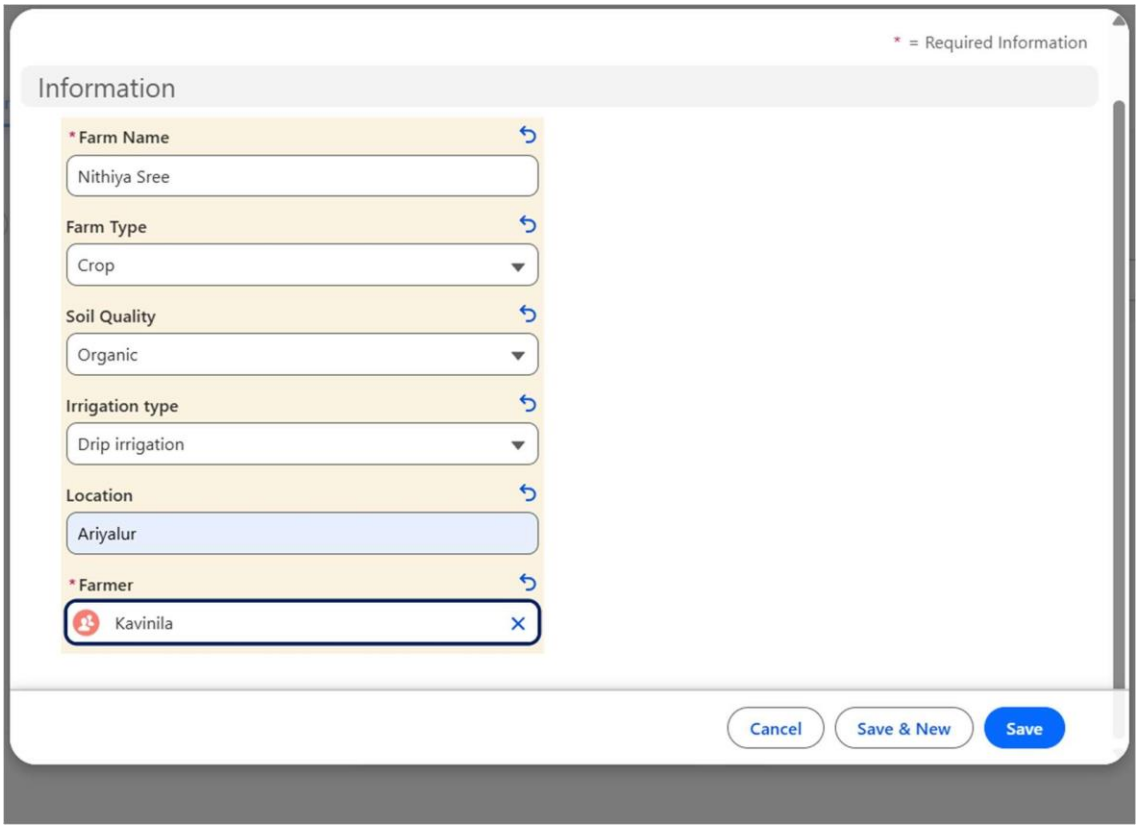
- Click on App Launcher on left side of screen.
- Search Farm Management & click on it.
- Click on Farmer Tab.
- Click new and fill details & Save



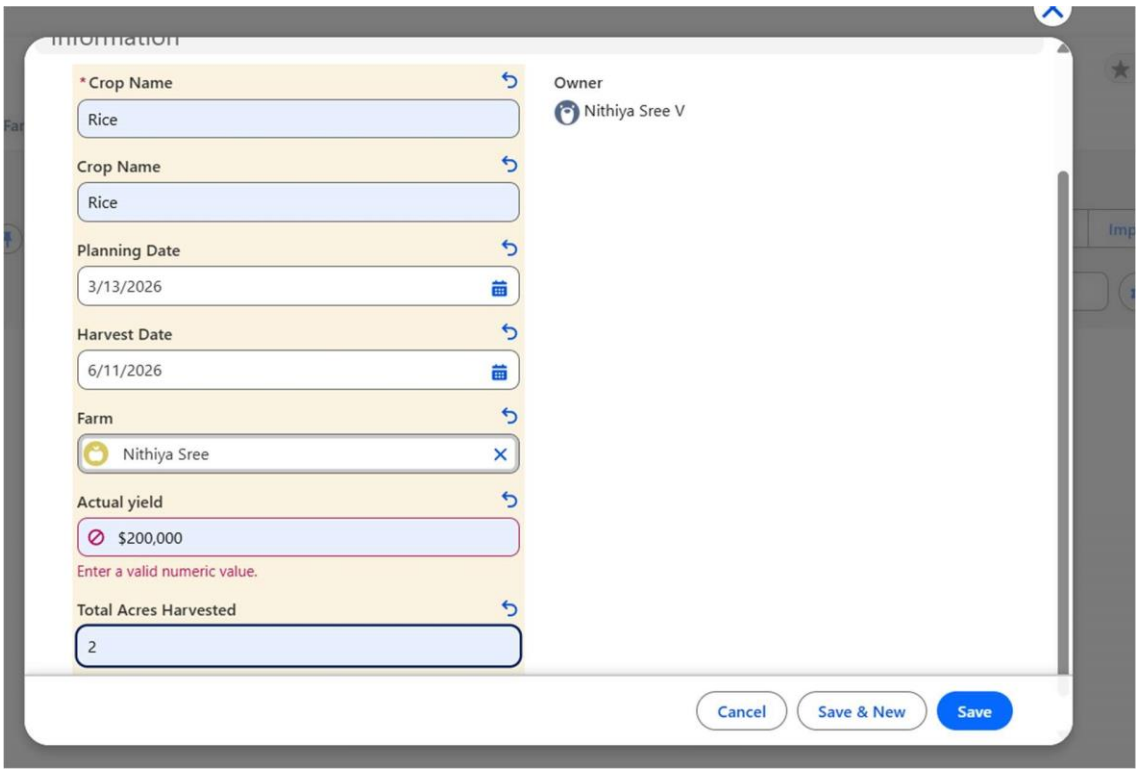
1.Adding Values to Farmers tab:



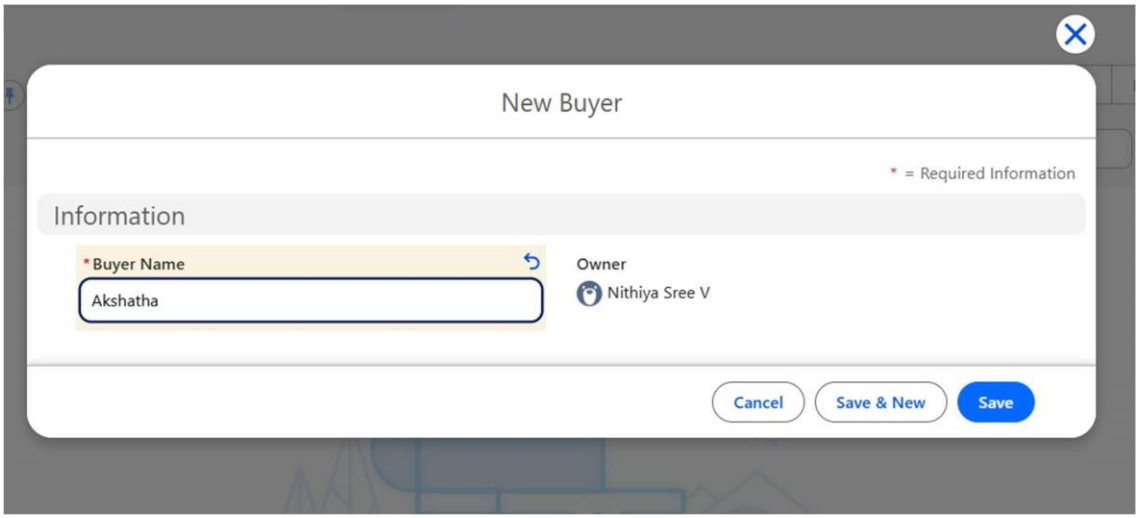
2.Adding values to Farm Tab:



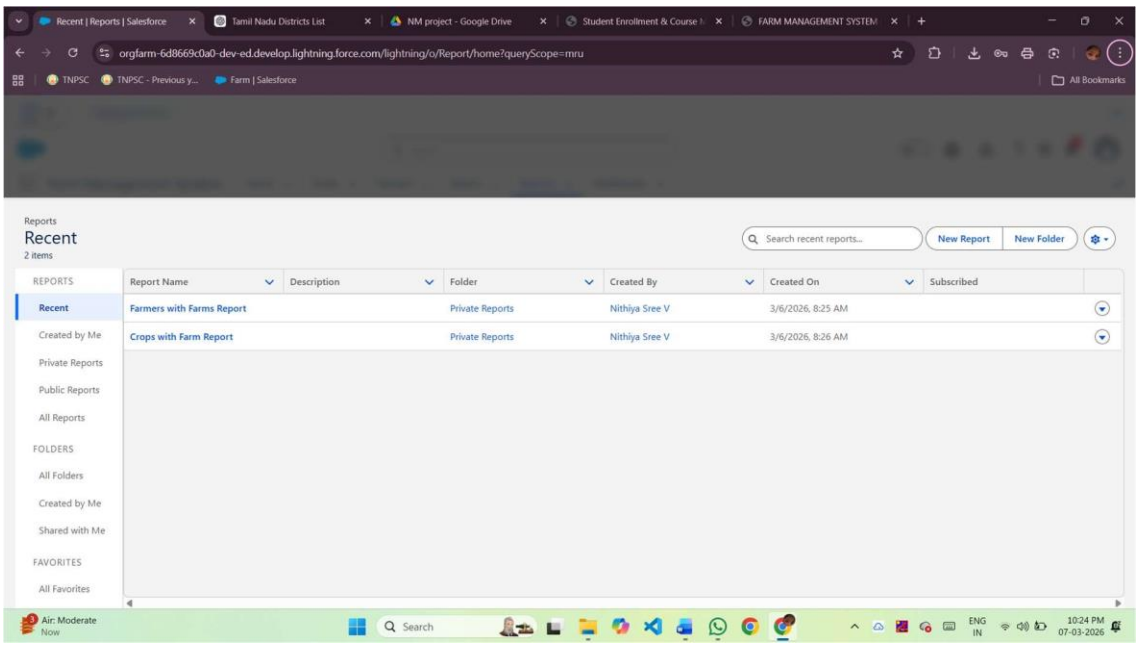
3.Adding Values to Crop tab:



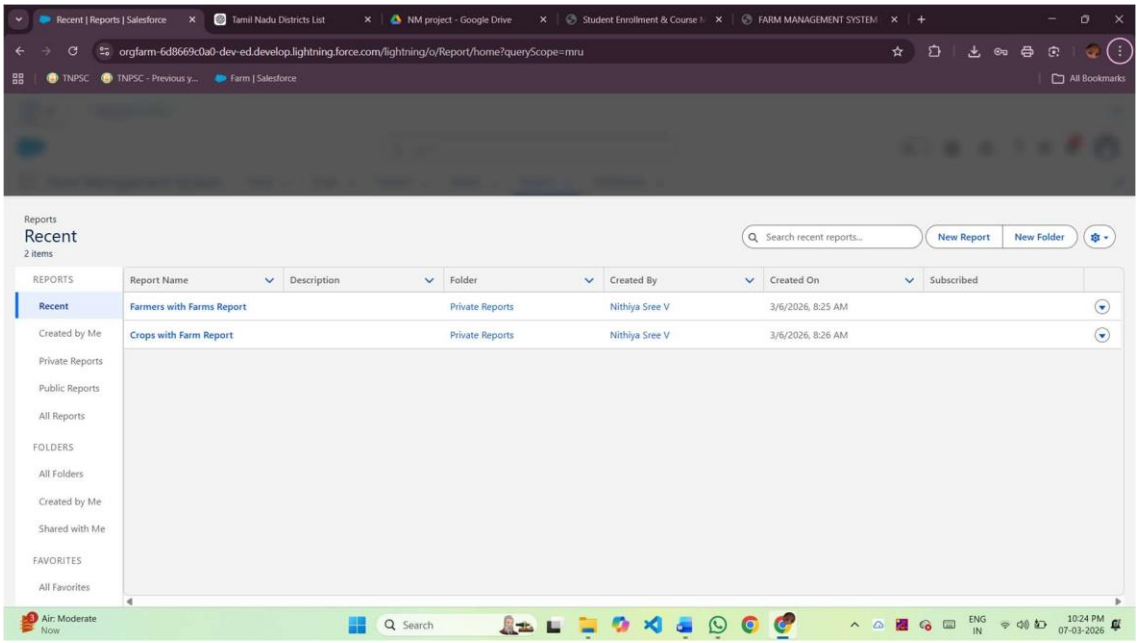
4.Adding Values to Buyer Tab:

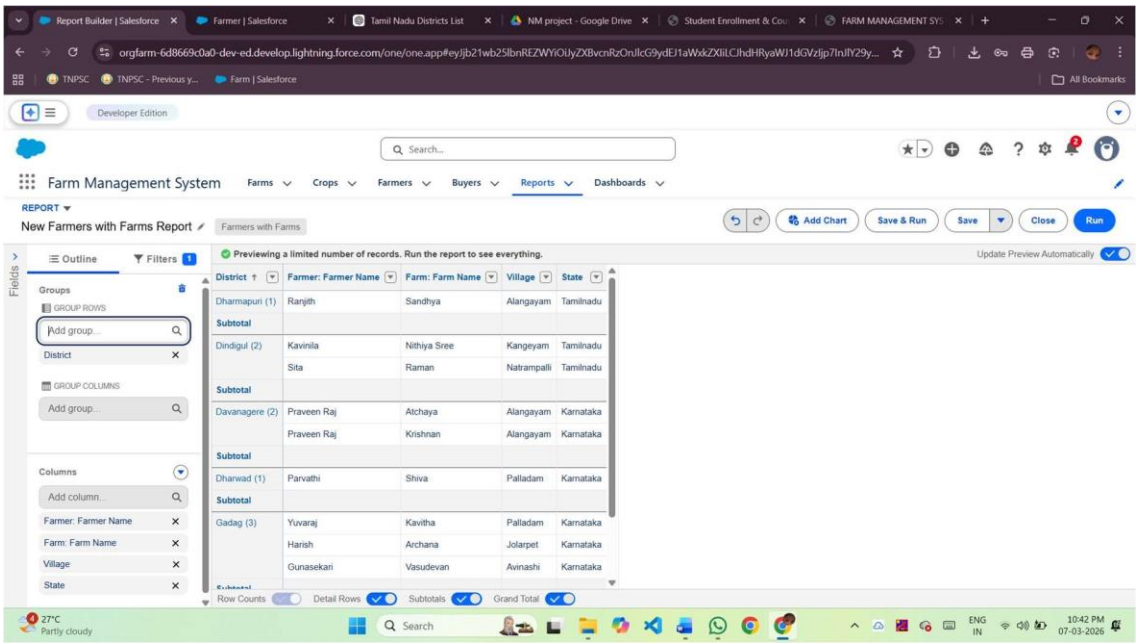


Milestone 11: Reports

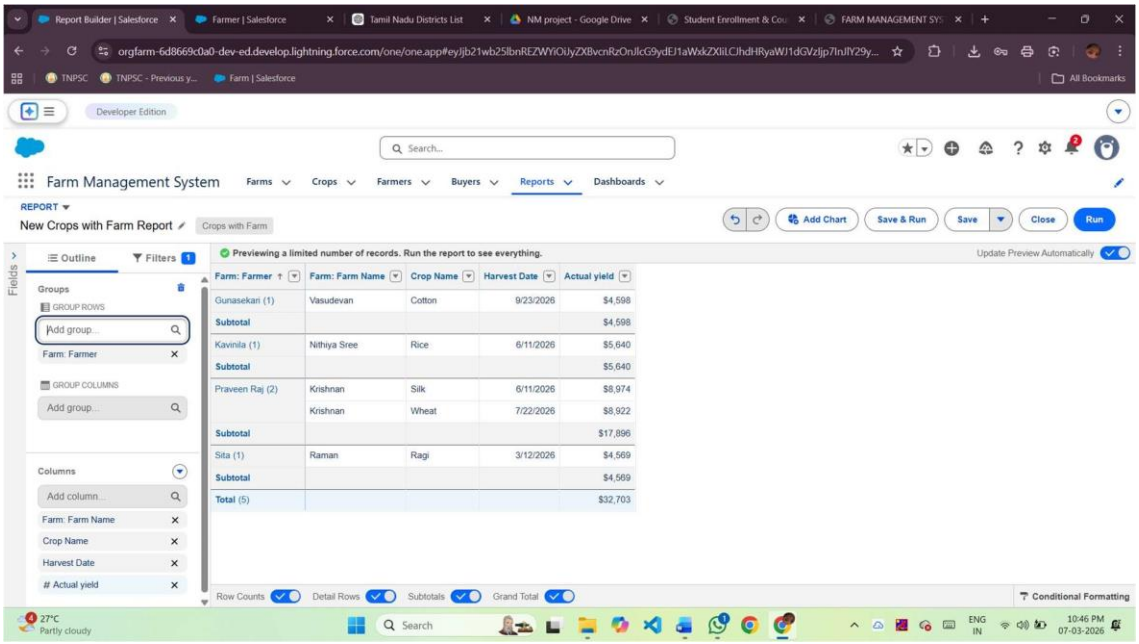


111.1 For Farmers with farms report:


Recent | Reports | Salesforce
Tamil Nadu Districts List
NM project - Google Drive
Student Enrollment & Course
FARM MANAGEMENT SYSTEM
orgfarm-6d8669c0a0-dev-ed.develop.lightning.force.com/lightning/o/Report/home?queryScope=mru
Reports
Recent
2 items
Search recent reports...
New Report
New Folder
REPORTS
Recent
Created by Me
Private Reports
Public Reports
All Reports
FOLDERS
All Folders
Created by Me
Shared with Me
FAVORITES
All Favorites
Report Name
Description
Folder
Created By
Created On
Subscribed
Farmers with Farms Report
Private Reports
Nithiya Sree V
3/6/2026, 8:25 AM
Crops with Farm Report
Private Reports
Nithiya Sree V
3/6/2026, 8:26 AM

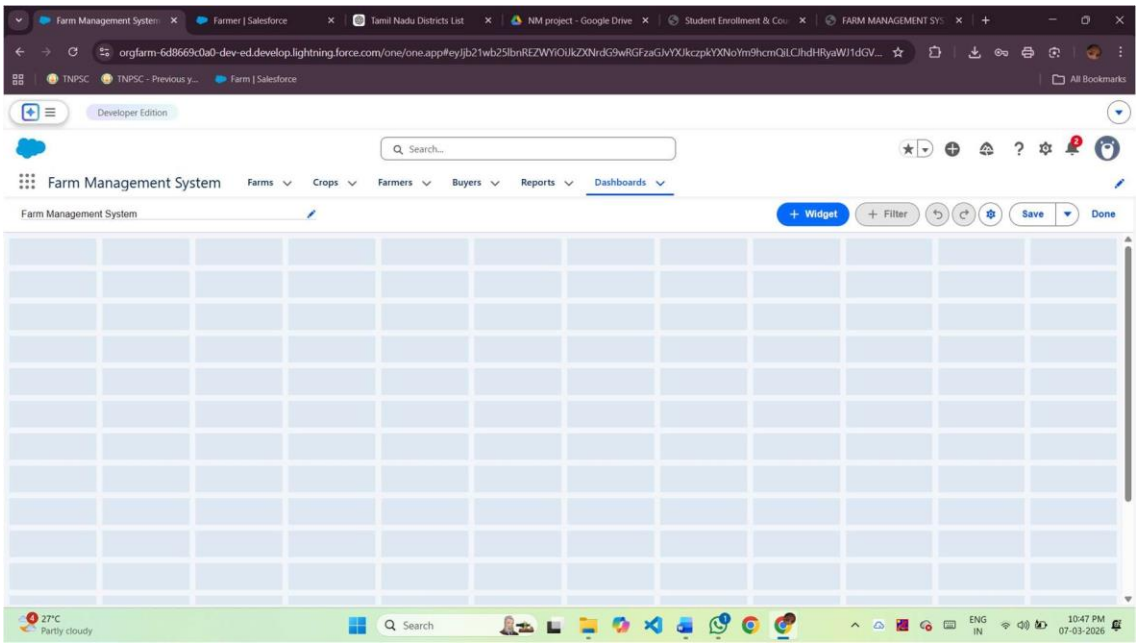

Report Builder | Salesforce
Farmer | Salesforce
Tamil Nadu Districts List
NM project - Google Drive
Student Enrollment & Cou
FARM MANAGEMENT SYS
orgfarm-6d8669c0a0-dev-ed.develop.lightning.force.com/one/one.app#eyJjb21wb25lbnREZWYiOiJyZXBvcnRzOnJlcG9ydEJ1aWxkZXliLCJhdHRyaWJ1dGVzIjp7InJlY29y...
Developer Edition
Farm Management System
Farms
Crops
Farmers
Buyers
Reports
Dashboards
REPORT
New Farmers with Farms Report
Farmers with Farms
Add Chart
Save & Run
Save
Close
Run
Outline
Filters
Previewing a limited number of records. Run the report to see everything.
Update Preview Automatically
Fields
Groups
GROUP ROWS
Add group...
District
GROUP COLUMNS
Add group...
Columns
Add column...
Farmer: Farmer Name
Farm: Farm Name
Village
State
District
Farmer: Farmer Name
Farm: Farm Name
Village
State
Dharmapuri (1)
Ranjith
Sandhya
Alangayam
Tamilnadu
Subtotal
Dindigul (2)
Kavinila
Nithiya Sree
Kangeyam
Tamilnadu
Sita
Raman
Natrampalli
Tamilnadu
Subtotal
Davanagere (2)
Praveen Raj
Atchaya
Alangayam
Karnataka
Praveen Raj
Krishnan
Alangayam
Karnataka
Subtotal
Dharwad (1)
Parvathi
Shiva
Palladam
Karnataka
Subtotal
Gadag (3)
Yuvaraj
Kavitha
Palladam
Karnataka
Harish
Archana
Jolarpet
Karnataka
Gunasekari
Vasudevan
Avinashi
Karnataka
Row Counts
Detail Rows
Subtotals
Grand Total

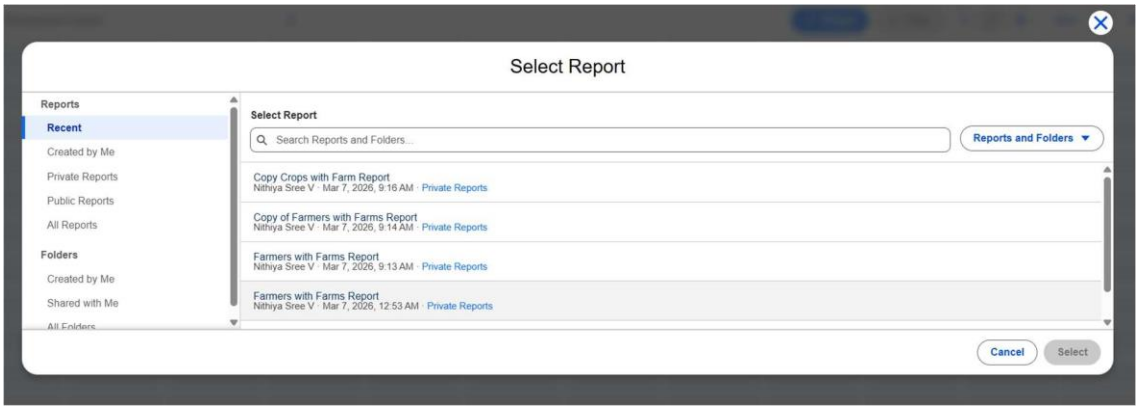
11.2 For Farmers with farms report:

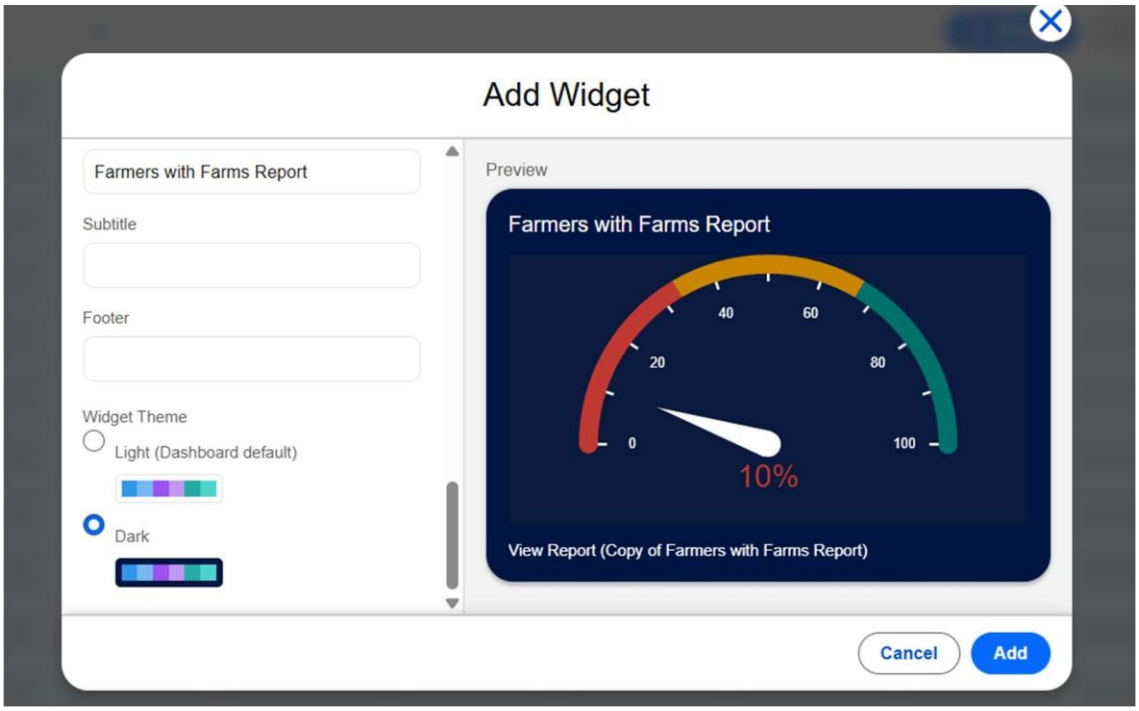


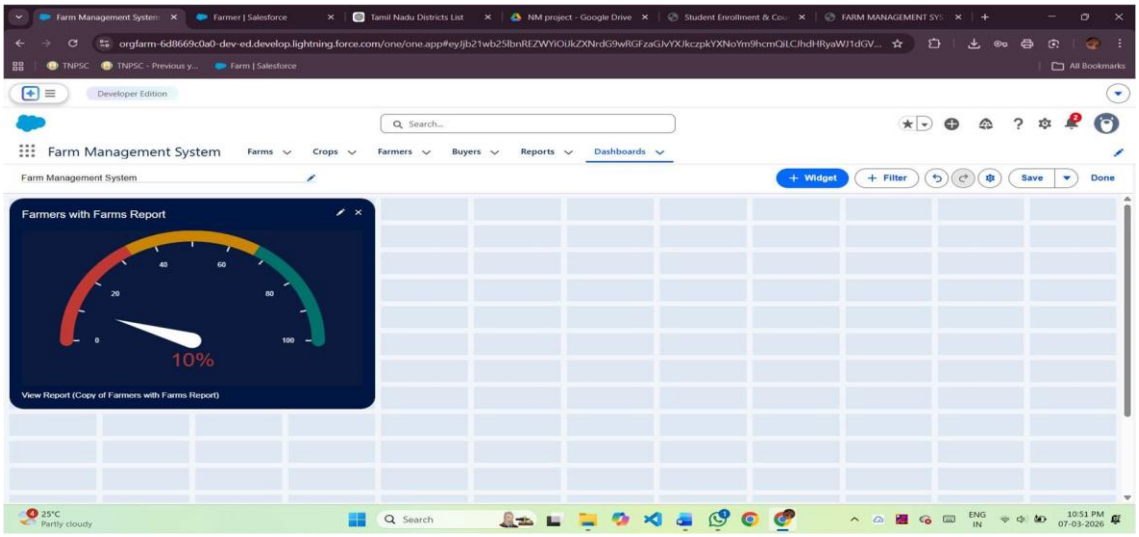
Milestone 12: Dashboard

Dashboard for Farmer with farm report:

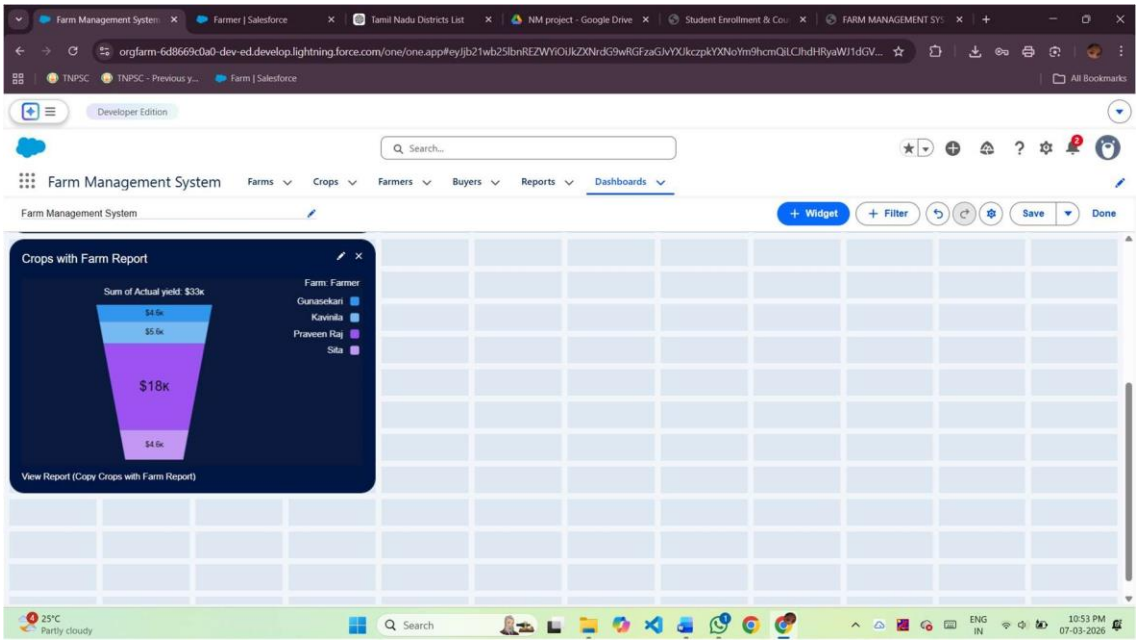



Select Report
Reports
Recent
Created by Me
Private Reports
Public Reports
All Reports
Folders
Created by Me
Shared with Me
Select Report
Search Reports and Folders...
Reports and Folders
Copy Crops with Farm Report
Nithiya Sree V · Mar 7, 2026, 9:16 AM · Private Reports
Copy of Farmers with Farms Report
Nithiya Sree V · Mar 7, 2026, 9:14 AM · Private Reports
Farmers with Farms Report
Nithiya Sree V · Mar 7, 2026, 9:13 AM · Private Reports
Farmers with Farms Report
Nithiya Sree V · Mar 7, 2026, 12:53 AM · Private Reports
Cancel
Select

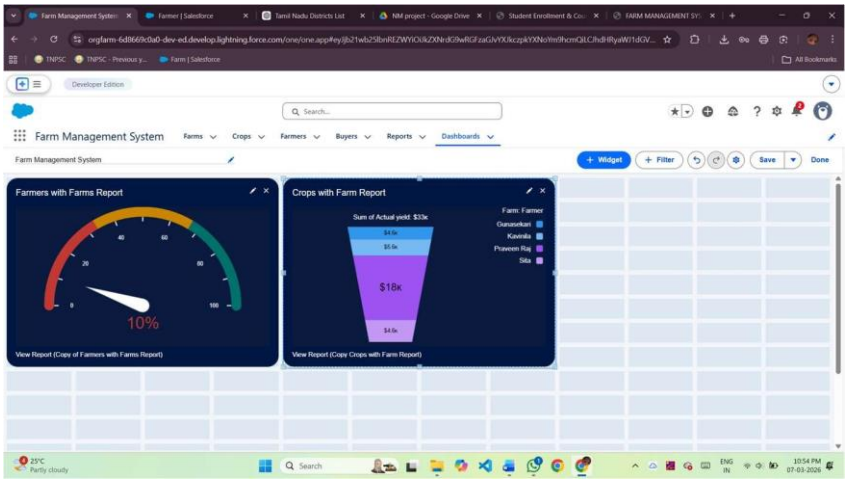

Add Widget
Farmers with Farms Report
Subtitle
Footer
Widget Theme
Light (Dashboard default)
Dark
Preview
Farmers with Farms Report
0
20
40
60
80
100
10%
View Report (Copy of Farmers with Farms Report)
Cancel
Add



Dashboard for Crops with farm objects:



Overall Dashboard:



FUNCTIONAL & PERFORMANCE TESTING

The testing phase was carried out on the Salesforce-based Farm Management System (FMS) to ensure that all modules operate correctly and meet the project requirements. Test cases were executed through the Salesforce interface, and screenshots were recorded to confirm the successful implementation of system features.

The following functionalities were evaluated:

- Successful creation and updating of Farmer, Farm, Crop, and Buyer records
- Verification of mandatory fields and input constraints to prevent invalid data entry
- Validation of object relationships, ensuring crops are correctly linked to farms and farmers ● Testing user access permissions to confirm that each role can view or edit only authorized data
- Checking data sharing configurations to ensure secure information visibility among users ● Generating analytical reports and visual dashboards for monitoring crop listings and buyer activities

In addition to functional verification, performance evaluation was conducted to observe system responsiveness during repeated data entry, record updates, and report generation. The results confirmed that the platform maintains stable performance, quick processing, and consistent reliability even when multiple operations are executed within the system.

ADVANTAGES & DISADVANTAGES

Advantages

- Centralized Farm Information – All farm, crop, farmer, and buyer details are stored in a single system for easy access.
- Efficient Data Management – Reduces manual record keeping and simplifies farm data organization.
- Improved Data Accuracy – Validation rules help prevent incorrect or incomplete information.
- Better Decision Support – Reports and dashboards provide insights into crop availability and farm activities.
- Easy Buyer–Farmer Interaction – Buyers can quickly view available crops and send purchase requests.

Disadvantages

- Internet Dependency – The system requires a stable internet connection to access Salesforce services.
- Platform Dependency – The system relies on the Salesforce platform for hosting and management.
- Limited Offline Functionality – Users cannot access or update data when the system is offline.

CONCLUSION

- The Farm Management System (FMS) developed using the Salesforce platform provides an efficient and organized solution for managing agricultural data. The system integrates key modules such as Farmer, Farm, Crop, and Buyer management, enabling farmers to maintain farm records and allowing buyers to easily view available crops.
- By centralizing farm-related information, the system reduces manual work, improves data accuracy, and simplifies the process of managing crop listings and buyer interactions. Features such as automation, role-based security, reports, and dashboards help in monitoring farm activities and analyzing crop availability effectively.
- Overall, the Farm Management System improves the efficiency of farm data management and supports better coordination between farmers and buyers. The system can also be expanded in the future with additional features such as weather

data integration, mobile access, and advanced analytics to further enhance agricultural management.

FUTURE SCOPE

- The Farm Management System (FMS) can be further enhanced by adding advanced features and integrations to improve agricultural management and farmer–buyer interaction.
- ● Weather Integration – Integrating real-time weather data to help farmers make better crop planning decisions.
- Mobile Application Support – Developing a mobile version of the system so farmers can manage farm data and crop listings easily using smartphones. ● Online Payment Integration – Allowing buyers to complete crop purchases through secure digital payment methods.
- Crop Recommendation System – Suggesting suitable crops based on soil type, season, and rainfall data.
- Advanced Analytics – Using data analytics to predict crop demand, production trends, and farm performance.
- Multi-language Support – Providing regional language options to make the system accessible for more farmers.
- These improvements will help make the system more scalable, intelligent, and user-friendly, supporting better farm management and agricultural productivity in the future.

APPENDIX

- The appendix section provides additional supporting materials and information related to the development and implementation of the Farm Management System (FMS).

 Appendix A – System Screenshots

 Screenshots captured from the Salesforce platform showing the implementation of key modules such as Farm, Crop, Farmer, and Buyer objects, page layouts, reports, dashboards, and automation flows.

 Appendix B – Data Model

 Details of the custom objects and relationships used in the system, including the connections between Farm, Crop, Farmer, and Buyer records.

 Appendix C – Testing Evidence

 Screenshots and records of functional testing, including validation checks, rolebased access testing, and report generation verification.

 Appendix D – Project Artifacts

 Documentation related to Agile sprint planning, product backlog, sprint schedules, and user stories used during project development.

 Appendix E – Tools and Technologies

- List of tools and technologies used for system development, including the Salesforce platform, automation flows, reporting tools, and security configurations.
- These materials support the implementation and validation of the Farm Management System and provide additional reference information for the project.

THANK YOU